

Let's learn about **satellites!**

Space mission and **satellite** design course

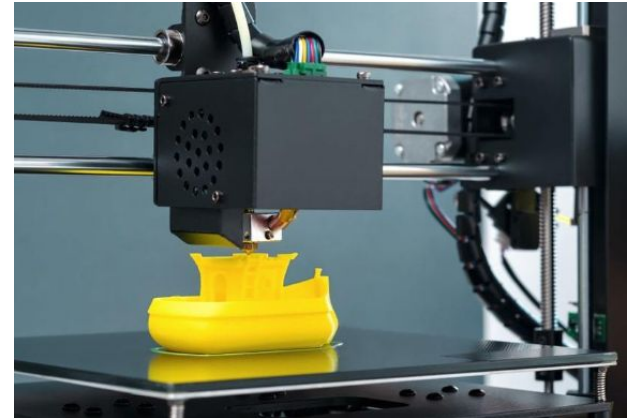
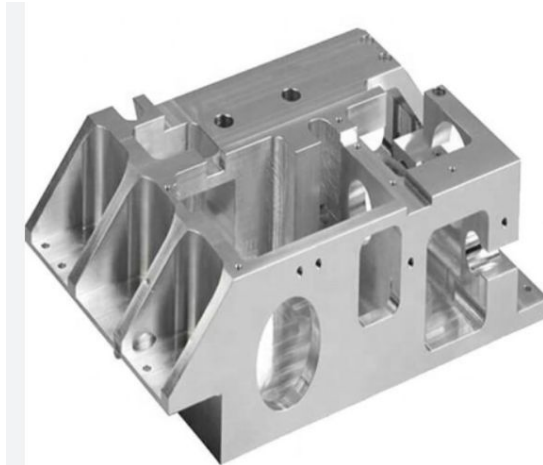
Structures

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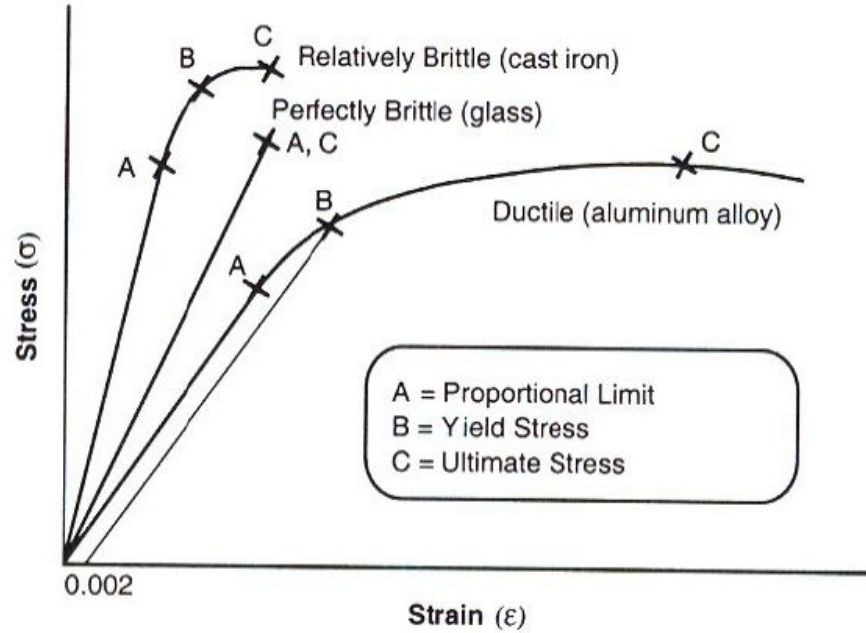
- Introduction on structural mechanics
 - Statics: mechanics of materials, material properties, stress state, loading conditions, material properties
 - Dynamics: vibration mechanics
- Subsystem description and requirements
 - Types of structures
 - Definition of requirements: flight loads (by flight phases and by type)
- Example of sizing of primary structure
 - Calculation of MoS **safety margins**, analytical verifications (sizing to strength and sizing to stiffness)
- Intro on FEM analysis

Mechanics of Material

- Isotropic materials are materials whose properties remain the same when tested in different directions.
- Isotropic materials differ from **anisotropic** materials, which display varying properties when tested in different directions. Common isotropic materials include glass, plastics, and metals. On the other hand, fiber-reinforced materials such as composites and natural materials such as wood tend to display anisotropic properties. These materials can demonstrate great strength when force is applied in the same direction as the fibers, and much less strength when force is applied in the opposite direction.



Mechanics of Material

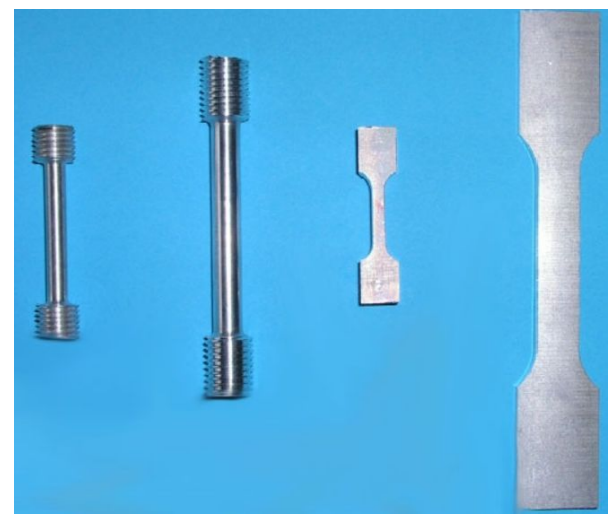


STRESS

$$\sigma = \frac{P}{A}$$

STRAIN

$$\epsilon = \frac{\Delta L}{L}$$



Mechanics of Material

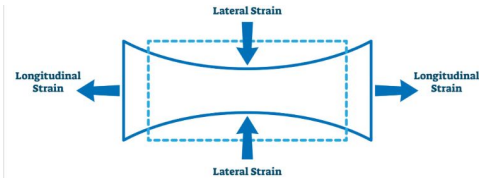
- Young's Modulus, also known as the modulus of elasticity, is a measure of a material's stiffness and its ability to withstand deformation under an applied force, quantifying the ratio of stress to strain within the material.

$$E = \frac{\sigma}{\epsilon} \quad \text{Pa}$$

Aluminum $\square$ 70 GPa
Steel $\square$ 210 GPa

- Poisson's ratio is a material property that defines how a material changes in width when it is stretched or compressed, indicating the ratio of transverse strain to axial strain.

$$\nu = - \frac{\epsilon_{m, \text{transv}}}{\epsilon_{m, \text{long}}}$$



Aluminum density $\square$ 2700 kg/m³
Steel density $\square$ 7800 kg/m³

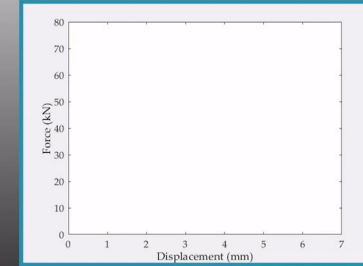
[LINK 1](#)

[LINK 2](#)

UNIAXIAL TENSILE TEST ON STAINLESS STEEL

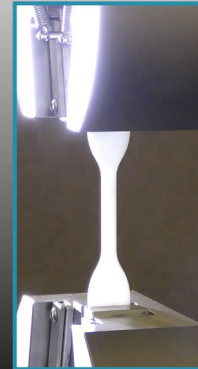


UNIAXIAL TENSILE TEST

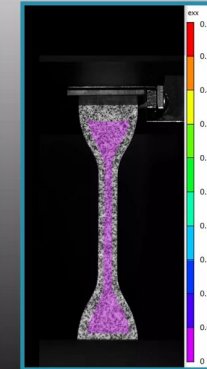


FORCE - DISPLACEMENT PLOT

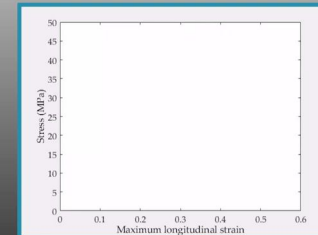
UNIAXIAL TENSILE TEST ON 3D PRINTED NYLON SLS



UNIAXIAL TENSILE TEST



LONGITUDINAL STRAIN

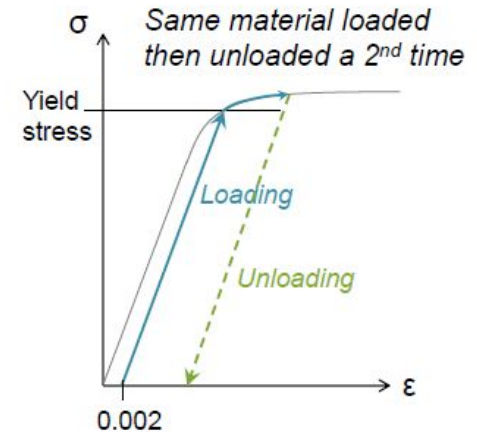
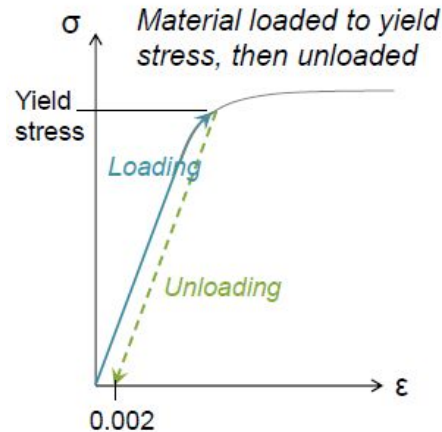
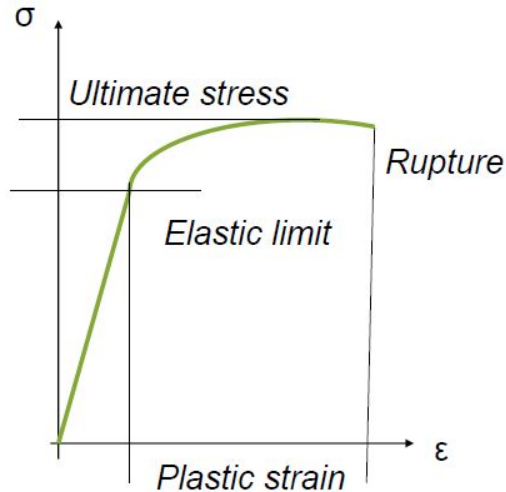


ENGINEERING STRESS - STRAIN PLOT

Mechanics of Material

- Tensile Strength σ_U
- Yield σ_Y (by convention, generates a residual strain of 0.2 %)

Yield strength is often the limit condition assumed for ductile materials



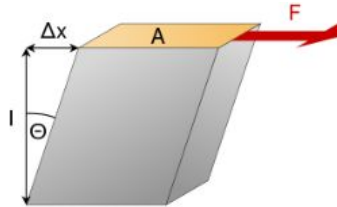
Mechanics of Material

Stress and strain

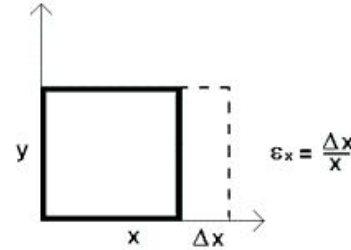
- Normal stress σ
- Shear stress τ
- Axial strain ϵ
- Shear strain γ

$$\sigma = \frac{P}{A}$$

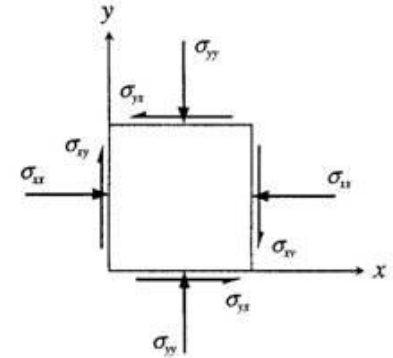
$$\epsilon = \frac{\Delta L}{L}$$



$$\gamma = \frac{\Delta x}{l}$$



$$G = \frac{\tau}{\gamma} \quad \text{SHEAR MODULUS}$$

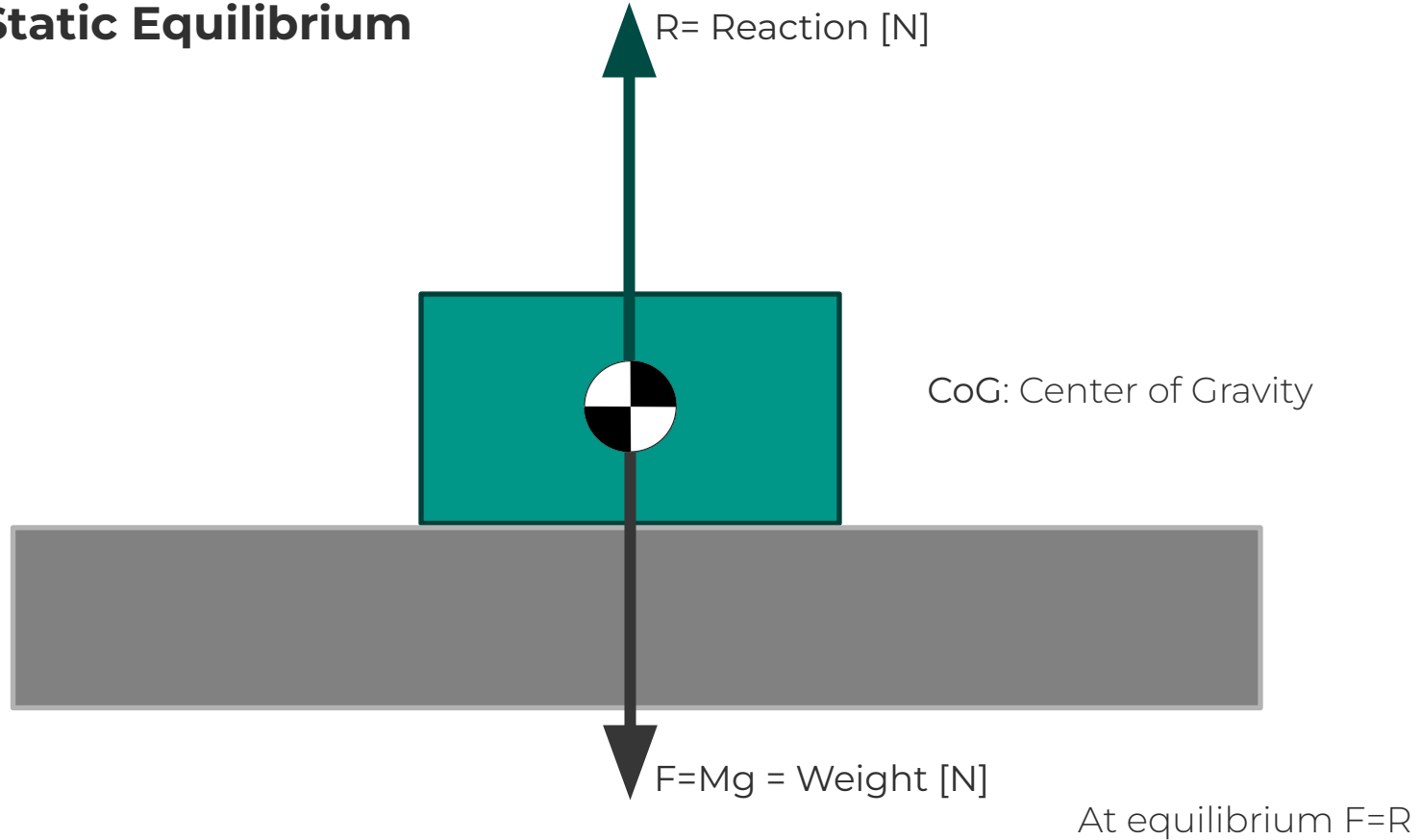


Resistance Criteria: Reduce the real stress state to a single-axial equivalent σ_{id} - von Mises (max distorting strain energy) Adopted for ductile materials

$$\sigma_{id} = \pm \sqrt{\sigma_x^2 + \sigma_y^2 - \sigma_x \sigma_y + 3\tau_{xy}^2}$$

Mechanics of Material

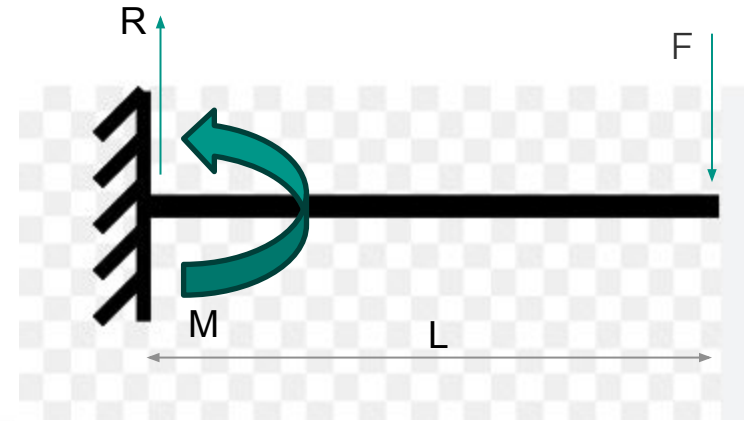
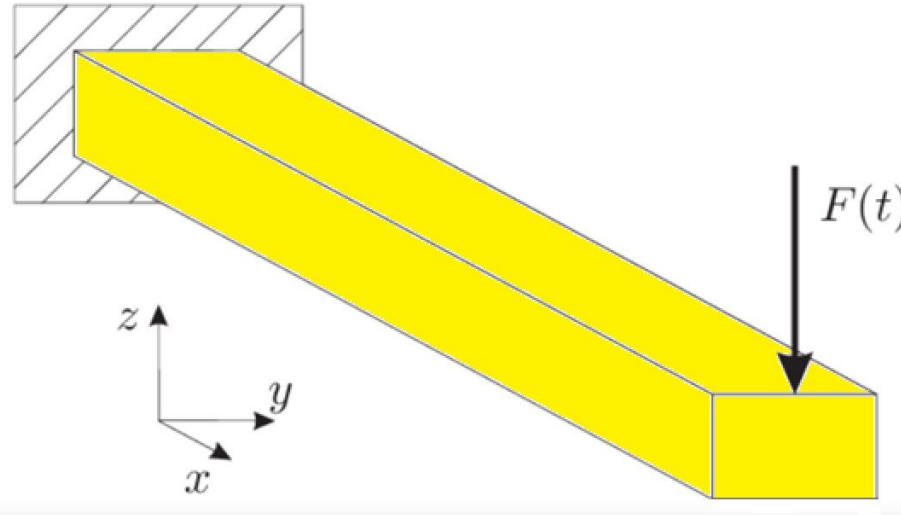
Static Equilibrium



Mechanics of Material

Static Equilibrium

Beam: a structure where one direction is predominant over the other two. Consider linear, isotropic, homogeneous material with a constant cross section.



$$R = F$$
$$M = F \times L$$

Mechanics of Material

Effect superposition: it is always possible to decompose the load into axial action, bending, shear, torque

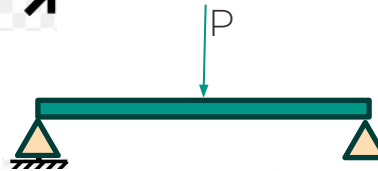
- Axial load



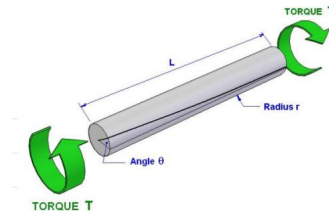
- Bending moment



- Shear load



- Torque



$$\sigma = \frac{P}{A}$$

A = area of the section

$$\sigma = \frac{PL}{I} \left(\frac{d}{2} \right)$$

I = Area moment of Inertia

$$\tau = \frac{P}{A}$$

r = radius

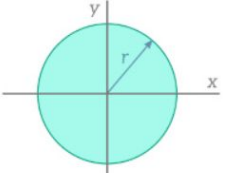
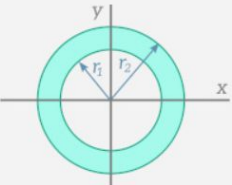
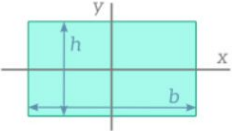
$$\tau = \frac{T}{J} r$$

J = Polar moment of Inertia

Mechanics of Material

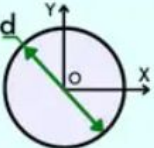
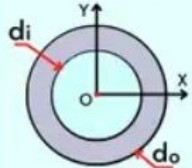
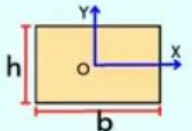
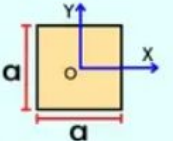
Area Moment of Inertia: The area moment of inertia, also called the second moment of area, is a parameter that defines how much resistance a shape (like the cross-section of a beam), has to bending because of its geometry.



Cross-Section	Area Moment of Inertia	
Filled Circle		$I_x = I_y = \frac{\pi}{4} r^4$
Hollow Circle		$I_x = I_y = \frac{\pi}{4} (r_2^4 - r_1^4)$
Rectangle		$I_x = \frac{bh^3}{12}$ $I_y = \frac{b^3h}{12}$

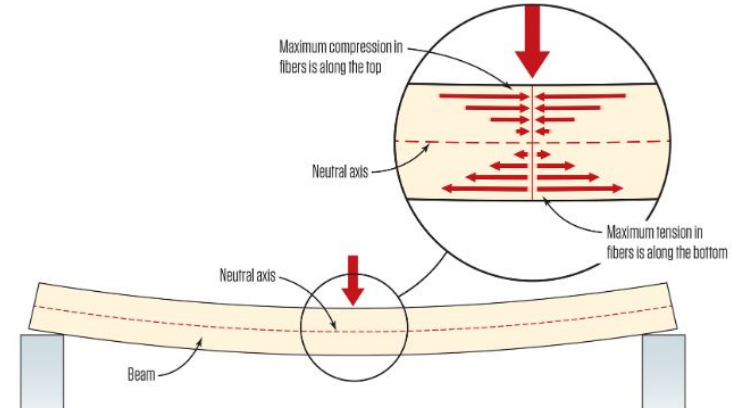
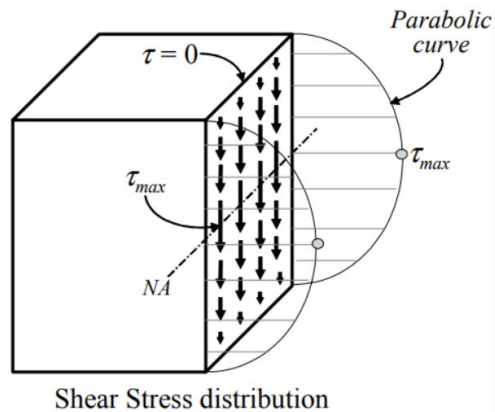
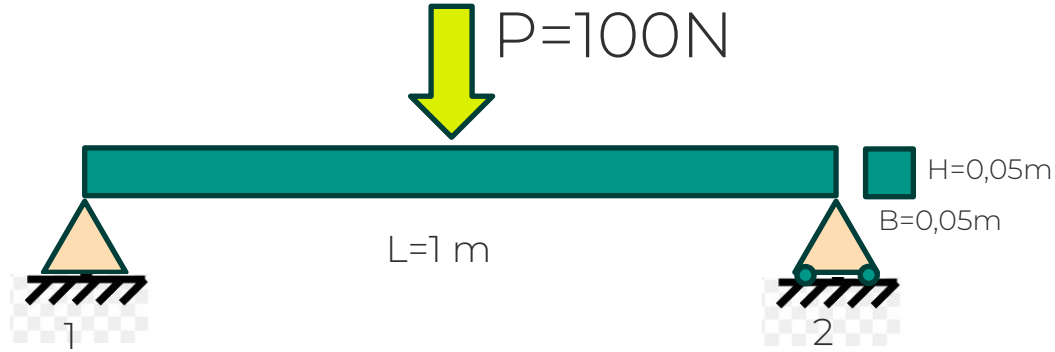
Mechanics of Material

...And to rotations

Shape	Polar moment of inertia, (J)
	$J_o = \frac{\pi}{32} \cdot d^4$
	$J_o = \frac{\pi}{32} \cdot [d_o^4 - d_i^4]$
	$J_o = \frac{1}{12} \cdot [bh^3 + hb^3]$
	$J_o = \frac{a^4}{6}$

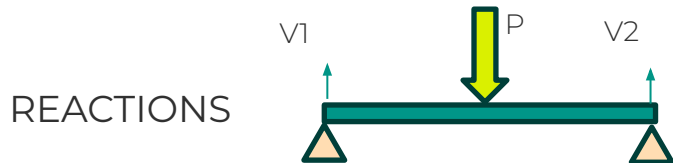
Mechanics of Material

BEAM EXAMPLE #1



Mechanics of Material

BEAM EXAMPLE



$$V_A = V_B = P/2 = 50 \text{ N}$$



$$\tau = \frac{P}{A} = \frac{100}{0,05 \cdot 0,05} = 40000 \text{ Pa}$$

$$I = \frac{bh^3}{12} = 5,21 \cdot 10^{-7} \text{ m}^4$$

$$\sigma = \frac{M}{I} \left(\frac{h}{2}\right) = \frac{\left(\frac{P}{2}\right)\left(\frac{L}{2}\right)}{I} \left(\frac{h}{2}\right) = 1200000 \text{ Pa}$$

EQUIVALENT STRESS

$$\sigma_{vm} = \sqrt{\sigma^2 + 3\tau^2} = 1202000 \text{ Pa}$$

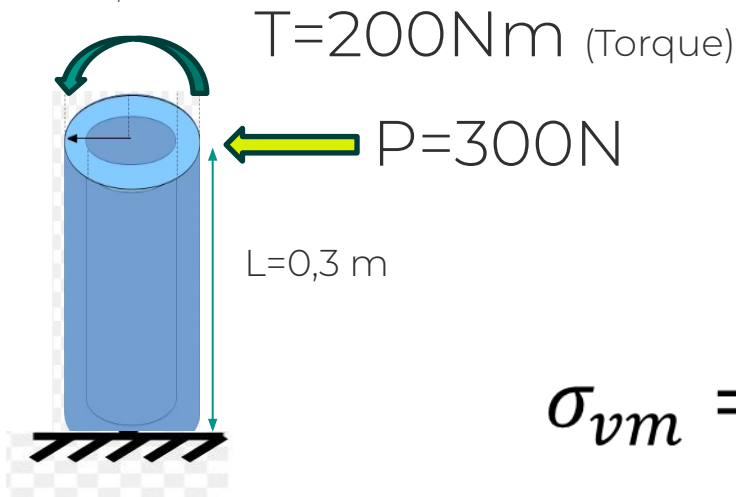
Mechanics of Material

BEAM EXAMPLE #2

The beam has a σ_y of 24 MPa

$D=0,1\text{ m}$

$d=0,098\text{ m}$



$$I = \frac{\pi (D^4 - d^4)}{64} = 3,81 \cdot 10^{-7} \text{ m}^4$$

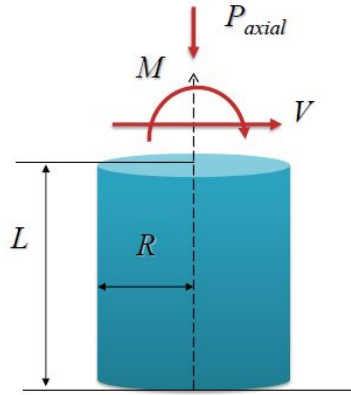
$$J = \frac{\pi (D^4 - d^4)}{32} = 7,62 \cdot 10^{-7} \text{ m}^4$$

$$\tau = \frac{T}{J} R = 13,12 \text{ MPa}$$

$$\sigma = \frac{M}{I} R = \frac{PL}{I} R = 11,8 \text{ MPa}$$

$$\sigma_{vm} = \sqrt{\sigma^2 + 3\tau^2} = 25,6 \text{ MPa}$$

Mechanics of Material



Uniaxial Element (Locked Beam): Compressive, Shear and Bending Load Conditions

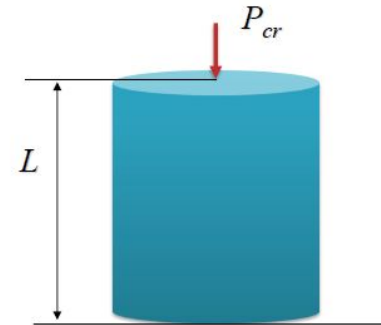
Equivalent axial load:

$$\sigma = \frac{P}{A} \pm \frac{MR}{I}$$

Compressive buckling (Eulerian critical load): "lateral deformation" that occurs in slender elements, depending on the type of end constraint

$$P_{cr} = A\sigma_{cr} = \frac{\pi^2 EI}{4L^2}$$

BUCKLING

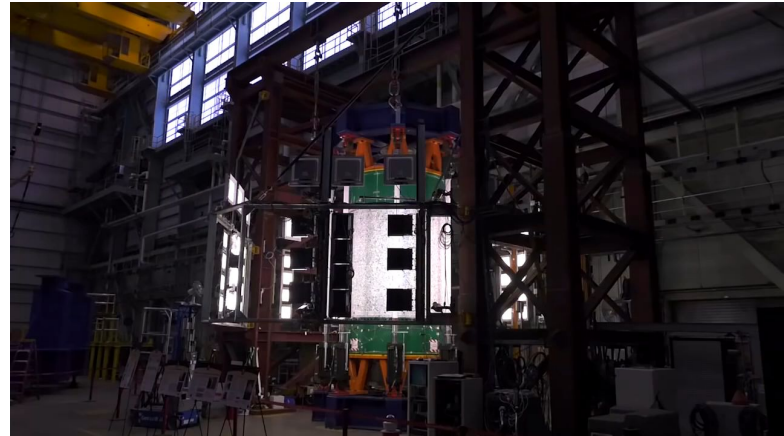


Mechanics of Material

BUCKLING 1

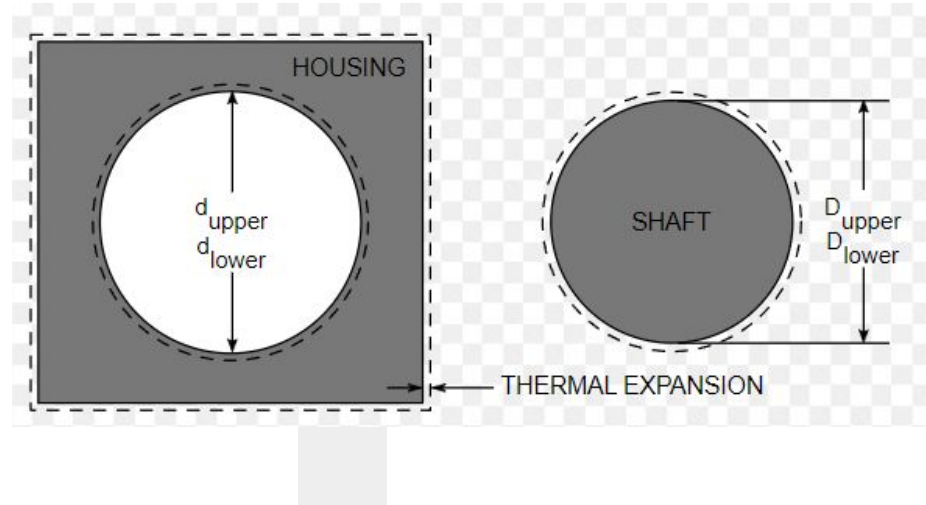
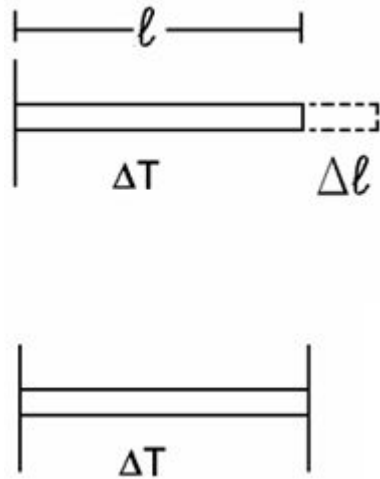


BUCKLING 2



Mechanics of Material

Mechanical stresses can arise due to temperature variations on components characterized by a certain Coefficient of Thermal Expansion - CTE α , but constrained in such a way as to prevent the corresponding expansion/contraction Δl .



Vibration

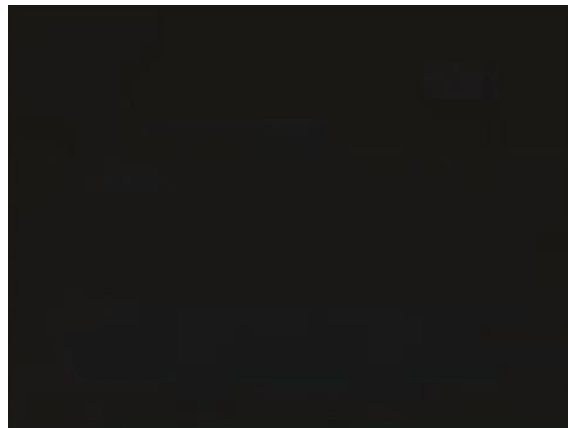
Vibration Mechanics – motion and deformations of a constrained structure under the action of variable external forces.

Natural frequency, is the frequency at which a system tends to oscillate in the absence of any driving force.

The motion pattern of a system oscillating at its natural frequency is called the normal mode (if all parts of the system move sinusoidally with that same frequency).

If the oscillating system is driven by an external force at the frequency at which the amplitude of its motion is greatest (close to a natural frequency of the system), this frequency is called resonant frequency.

RESONANCE



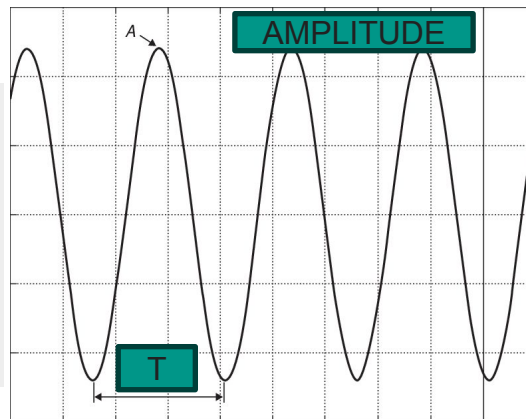
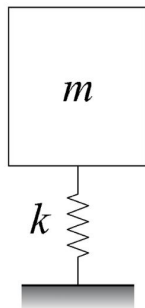
Credits: youtube mstkwn

Vibration

It is possible to approximate a complex structure (for instance a whole satellite) with very simple models to obtain preliminary results.

1 DOF: translating mass (can only move along x) constrained externally by an elastic element of stiffness k .

Mass – Spring system: free vibration



T = period

f = frequency

$f = 1/T$

$$\omega_0 = \sqrt{k/m} \quad [\text{rad/s}]$$

$$f_0 = \omega_0 / 2\pi \quad [\text{Hz}]$$

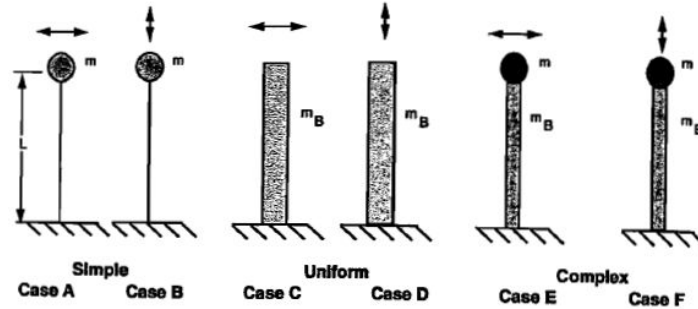
K : Stiffness $\square$ Hook's law $F = K \cdot x$

M : mass



Vibration

Natural frequency of beams



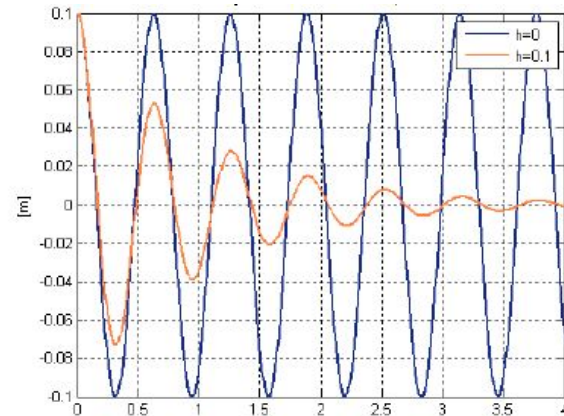
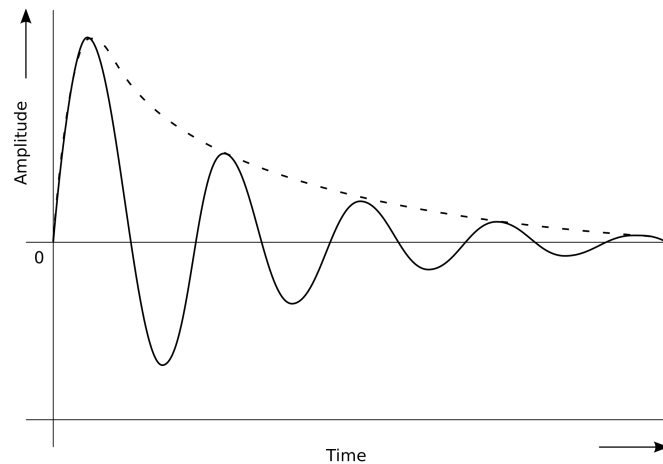
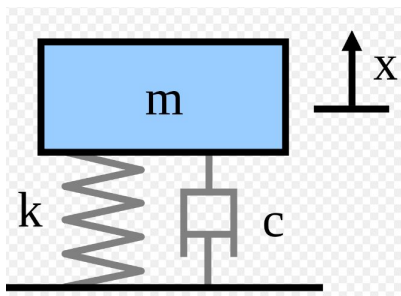
CASE A: Lateral Beam $m_B = 0 \quad T_p = m$ $\delta = 0.333 \left(\frac{mL^3}{EI} \right) ng$ $f_{nat} = 0.276 \sqrt{\frac{EI}{mL^3}}$	CASE C: Lateral Beam $m_B \neq 0 \quad T_p = 0$ $\delta = 0.125 \left(\frac{m_B L^3}{EI} \right) ng$ $f_{nat} = 0.560 \sqrt{\frac{EI}{m_B L^3}}$	CASE E: Lateral Beam $m_B \neq 0 \quad T_p = m$ $\delta = \delta_A + \delta_C$ $= \frac{L^3}{EI} (0.333 m + 0.125 m_B) ng$ $f_{nat} = 0.276 \sqrt{\frac{EI}{mL^3 + 0.236 m_B L^3}}$
CASE B: Axial Beam $m_B = 0 \quad T_p = m$ $\delta = \frac{mL}{AE} ng$ $f_{nat} = 0.160 \sqrt{\frac{AE}{mL}}$	CASE D: Axial Beam $m_B \neq 0 \quad T_p = 0$ $\delta = 0.5 \left(\frac{m_B L}{AE} \right) ng$ $f_{nat} = 0.250 \sqrt{\frac{AE}{m_B L}}$	CASE F: Axial Beam $m_B \neq 0 \quad T_p = m$ $\delta = \delta_B + \delta_D = \frac{L}{AE} (m + 0.5 m_B) ng$ $f_{nat} = 0.160 \sqrt{\frac{AE}{mL + 0.333 m_B L}}$

n = load factor
 g = gravitational acceleration
 m_B = mass of the beam (uniformly distributed)
 T_p = tip mass

I = area momentum of inertia of the beam's cross-section
 E = the modulus of the elasticity
 A = cross-sectional area of the beam

Vibration

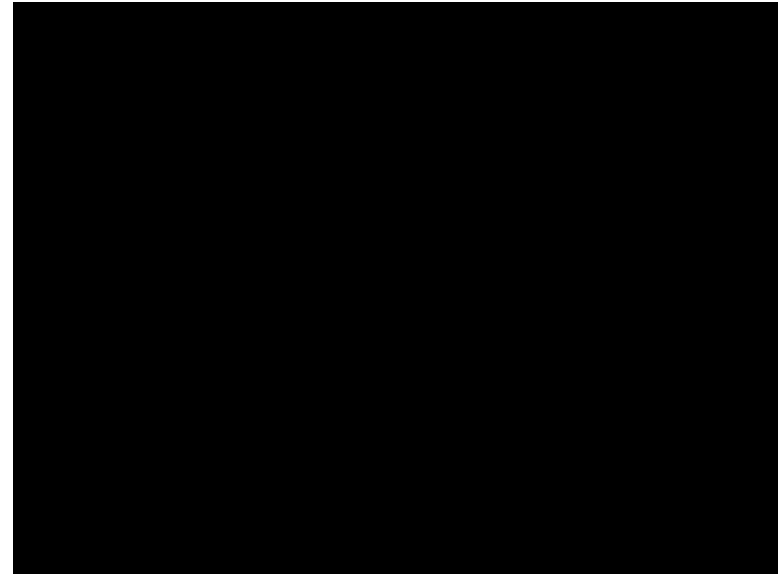
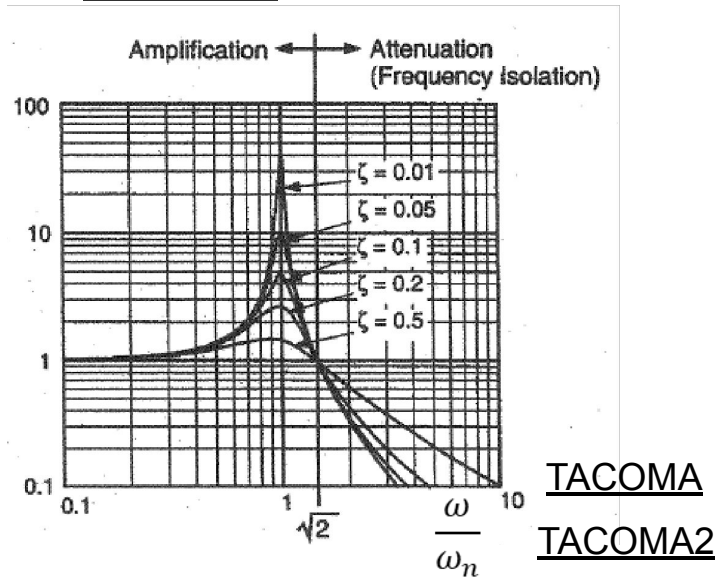
Mass – Spring - damper system: free vibration



Vibration

Forced vibration

Natural vibrations are different from forced vibrations which happen at the frequency of an applied force (forced frequency). If the forced frequency is equal to the natural frequency, the vibrations' amplitude increases manyfold. This phenomenon is known as resonance.



Subsystem Description

The structure performs the function of:

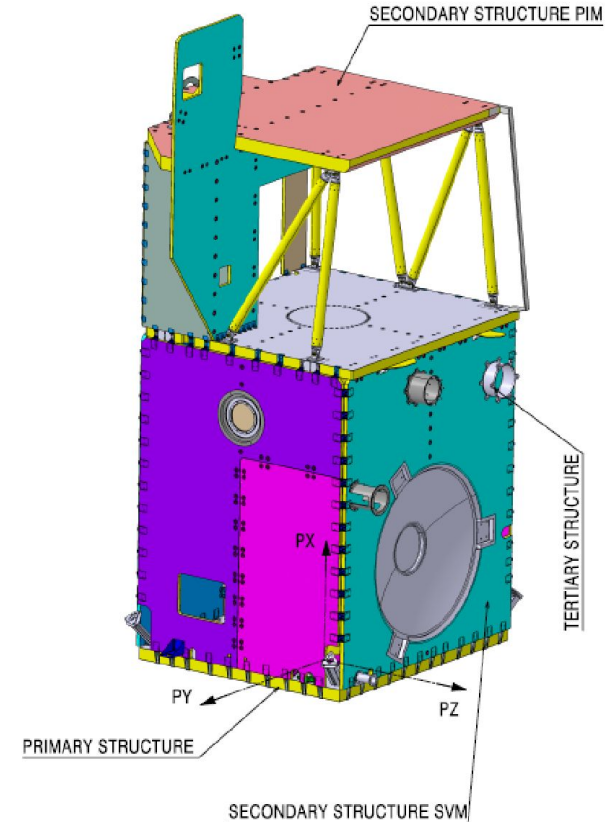
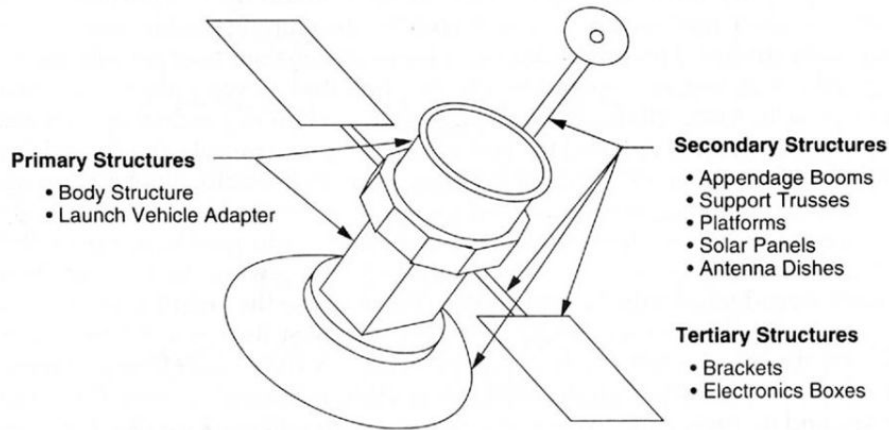
- provide mechanical support to the satellite, housing the payload and other on-board subsystems, and protecting them from mechanical (and thermal) loads;
- maintain the absolute and reciprocal configuration and position of the components;
- act as an interface to the launcher (through the payload adapter)

The structure subsystem consists of:

- **Primary structure:** carries the main loads on the satellite. Determines the fundamental natural frequency of the system.
- **Secondary structures:** support internal and external appendages and components (booms, solar panels, antennas, etc.)
- **Tertiary structures:** support individual components (brackets)

The mass of the components (structural elements, fixing systems) that constitute it must be as low as possible (launch costs).

Subsystem Description



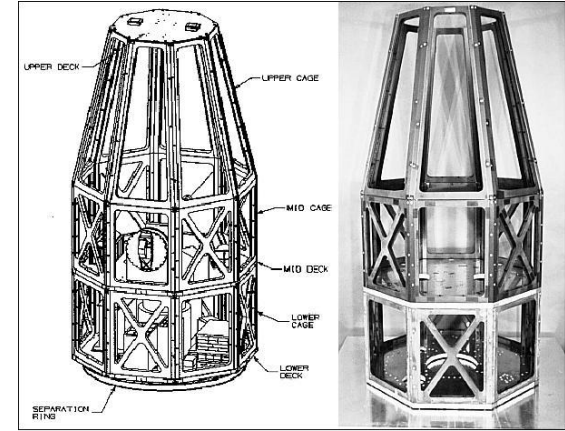
Subsystem Description

The main configurations for the primary structure adopted are:

Skin-frame: structure with reinforcements (stringers) oriented in the main load directions, surrounded by panels

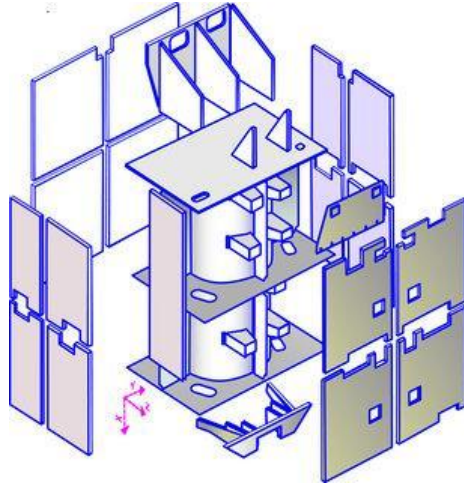
Monocoque cylinder: homogeneous shell structure (in aluminum alloy, titanium, composite) with cylindrical symmetry **without reinforcements**. Difficult accessibility; compression instability problems. Loads spread across the outer shell to maximize internal volume

Skin-stringer, semi-monocoque: shell reinforced by longitudinal elements (stringers) and circumferential elements (frames), fixed with mechanical connections (rivets) or welding.

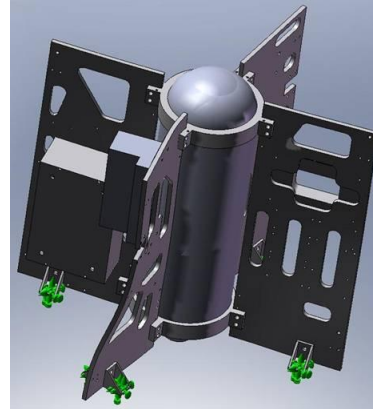


Subsystem Description

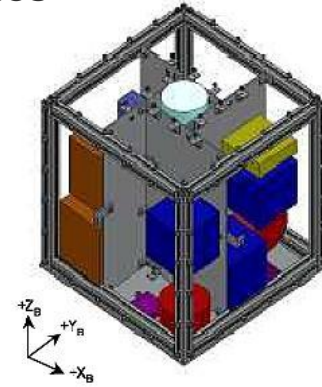
1 - Examples of structures and configurations for satellites



MetOp - Envisat



ShindaiSat

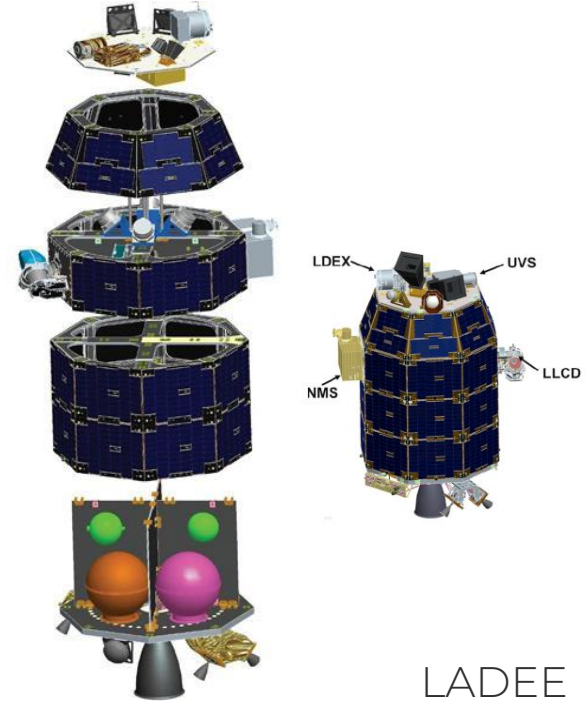


Subsystem Description

2 - Examples of structures and configurations for satellites



Proba-2



LADEE

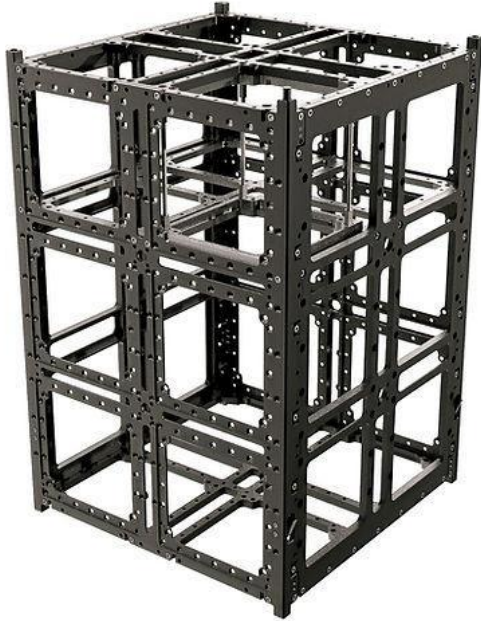
Subsystem Description

ISOGRID structures: typically used for rocket stages. This type of structure is characterized by the presence of stiffeners with a triangle pattern obtained on a plate or panel. The structure is extremely efficient from the point of view of stiffness and lightness, but is very expensive to create.



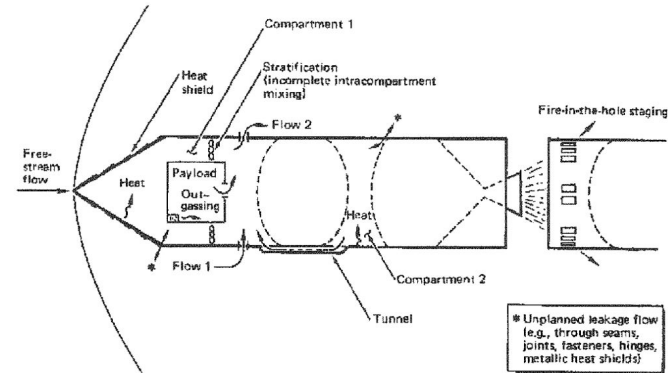
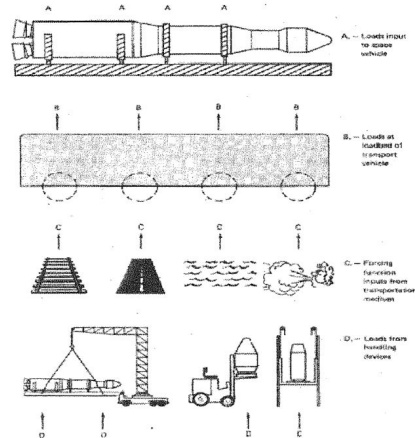
Subsystem Description

Examples of structures and configurations for nanosatellites – CubeSats



The SS Structures must carry out its function in a specific environment:

- **natural**: thermal, chemical, vacuum, radiation, debris conditions;
- **induced**: mechanical loads before launch (vibrations from transport, handling; testing), during launch and in orbit (thermoelastic loads and from humidity variations, accelerations from maneuvers, microvibrations from internal movements, docking, etc.), possible re-entry, descent and landing.



Requirements

Accelerations

- The parameter most commonly used (in the industry) to “define the motion of a mechanical system” is the acceleration
- Good reasons: accelerations are directly related to forces/stresses and “easy” to specify and measure
- In practice accelerations are used as a measure of the severity of the mechanical environment
- Criteria for equivalent structural damage (e.g. shock response spectra)

Note: failures usually happen in the largest stress areas, regardless if they are the largest acceleration areas!

□ Important consequences: Need for considering the “actual” (e.g. “test” or “flight”) boundary conditions

Requirements

Typical Requirements for Spacecraft Structures

- Strength
- Stiffness
- Structural life
- Damping
- Mass Properties
- Dynamic Envelope
- Positional Stability
- Mechanical Interface

Basic requirement: the structure shall support the payload and spacecraft subsystems with enough strength and stiffness to preclude any failure (rupture, collapse, or detrimental deformation) that may keep them from working successfully.



Mechanical loads

Mechanical loads are caused by:

- Transportation
- Rocket Motor Ignition Overpressure
- Lift-off Loads
- Engine/Motor Generated Acoustic Loads
- Engine/Motor Thrust Transients
- Pogo Instability, Solid Motor Pressure Oscillations
- Wind and Turbulence, Aerodynamic Sources
- Liquid Sloshing in Tanks
- Stage and Fairing Separation Loads
- Pyrotechnic Induced Loads
- Manoeuvring Loads
- Flight Operations, Onboard Equipment Operation

Mechanical loads

The main structural loads are those that arise during the **launch** induced by the launcher on the satellite, they can be grouped by type, level, time range or frequency.

A possible subdivision by type, also corresponding to that of tests for verifying requirements, is:

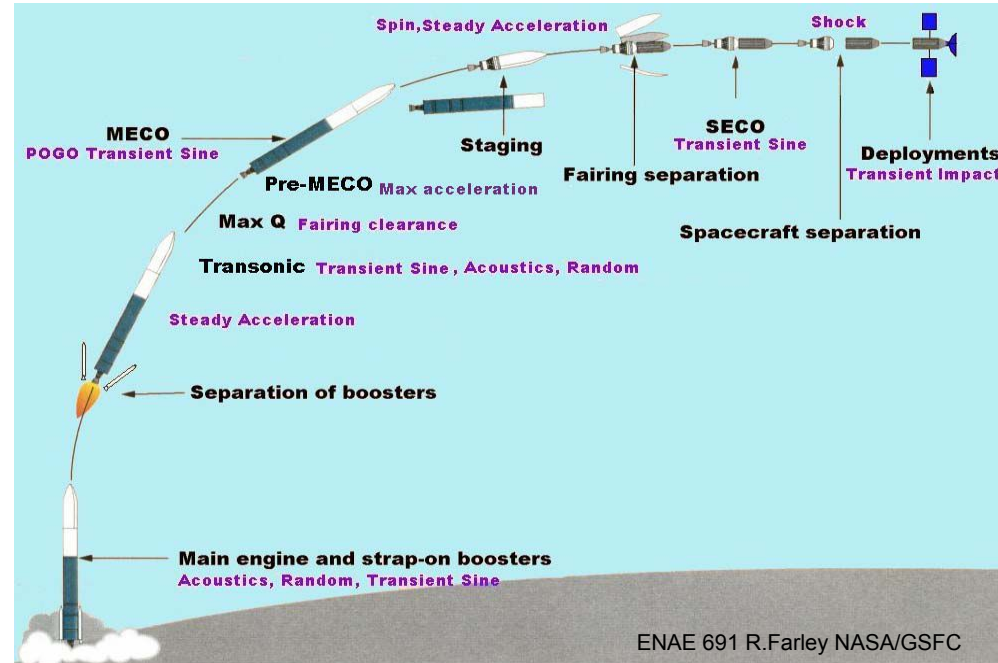
- Quasi/static (Quasi-Static Load QSL): engine thrust (and transient from staging), transport and gravity loads
- Low frequency dynamic Load (sine): stage motor cut-off, sloshing, maneuvering inertia, motor pressure oscillations (POGO);
- High frequency dynamics Load (random): ignition, thrust transient and engine-induced vibrations, aerodynamic loads, wind and turbulence;
- Acoustic Load: vibrations induced by pressure waves acting on large surfaces;
- Shock: strongly transient impulse loads from separations (pyrotechnics)

The limits of the "low" and "high" frequency ranges depend on the characteristics of the system: the limit between the two ranges is the first natural frequency of the satellite

Loads are transmitted to the S/C from the LV through the interface, acoustic loads are transmitted directly

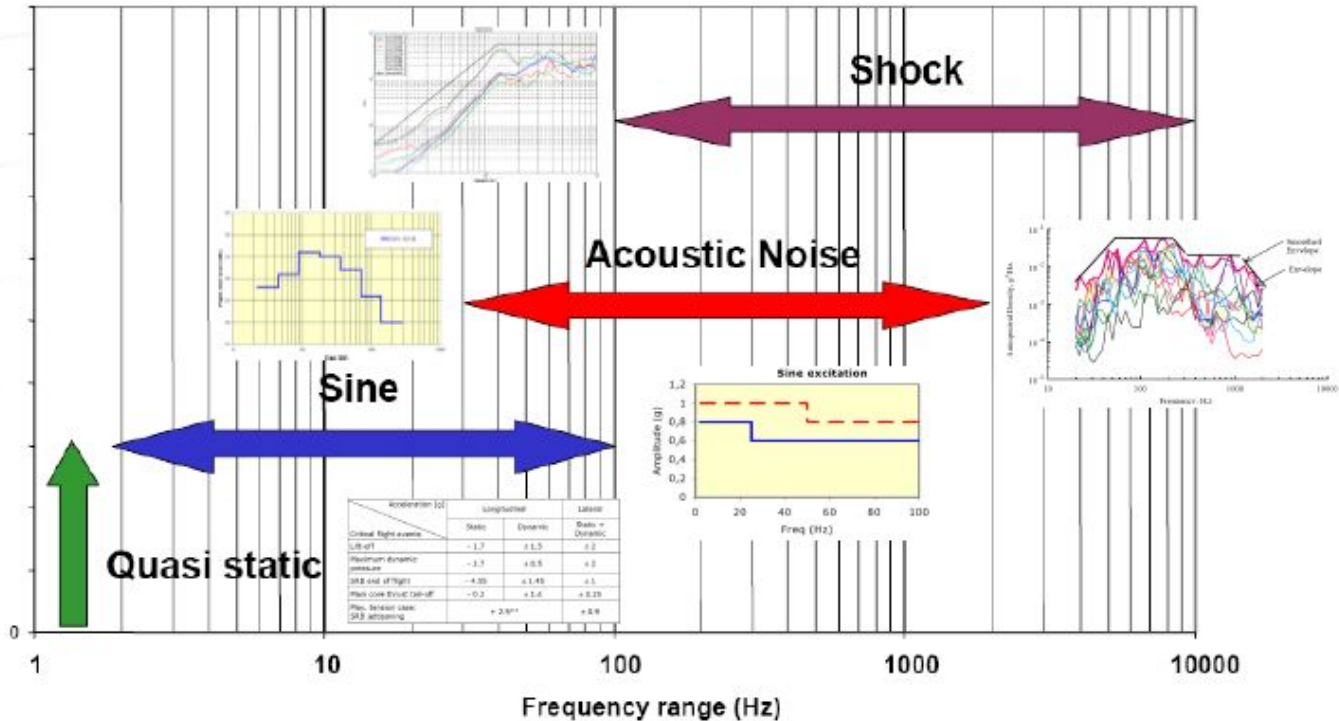
Mechanical loads

The main structural loads for the satellite induced by the launcher, grouped by mission phases:



Mechanical loads

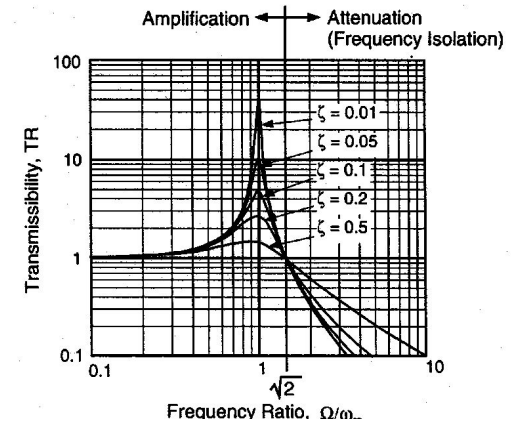
Structural loads on the satellite grouped by frequency range:



Stiffness Requirements

Launch Vehicle	In lateral axis	In longitudinal axis (thrust direction)
Dnepr	≥ 15 Hz	20 Hz - 45 Hz
Delta II 7320	> 12 Hz	> 35 Hz
H-IIA	> 10 Hz	> 30 Hz
PSLV	> 18 Hz	> 40 Hz
Soyuz BAI/KOU (spacecraft + adapter)	> 12 Hz	> 27 Hz
Soyuz BAI/KOU (spacecraft)	> 15 Hz	> 35 Hz
Zenit 2 SLB	> 6 Hz	> 20 Hz

Launchers prescribe limits on the axial and lateral natural frequencies of the S/C to avoid dynamic coupling phenomena, which can lead to high response amplifications

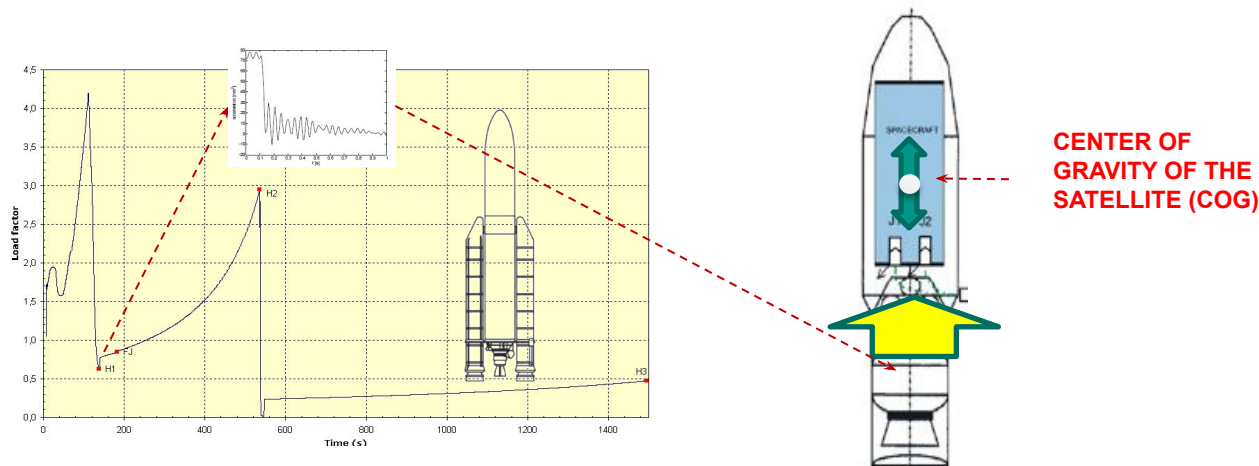


Amplification of the dynamic response of a damped SDOF system

Sizing

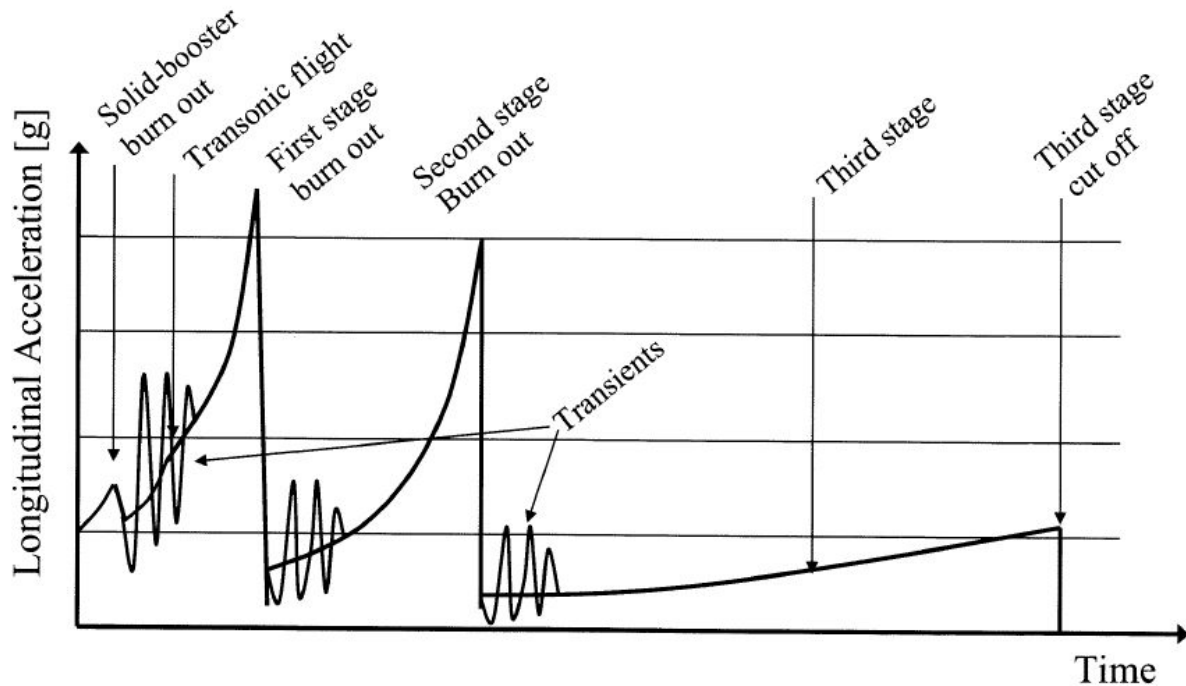
The Design Loads adopted for the first sizing of a satellite/component are the QSL Quasi-Static Loads: they are specified in the launcher's User Manual as accelerations (expressed in g) and arise from a combination of **low frequency** phenomena on the ground and in flight (thrust and engine ignition). They are applied to the **center of mass** of the satellite.

They are applicable only if the requirement on dynamic decoupling between SC and LV is satisfied (stiffness req.)



Sizing

Steady-state accelerations and low-frequency transients

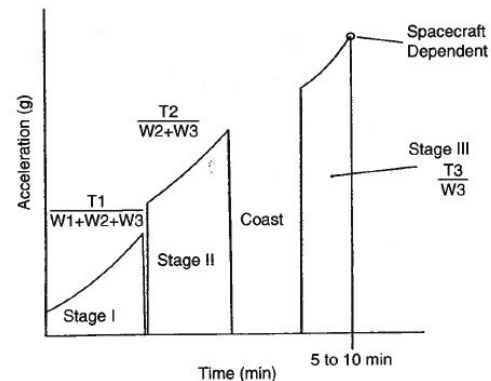
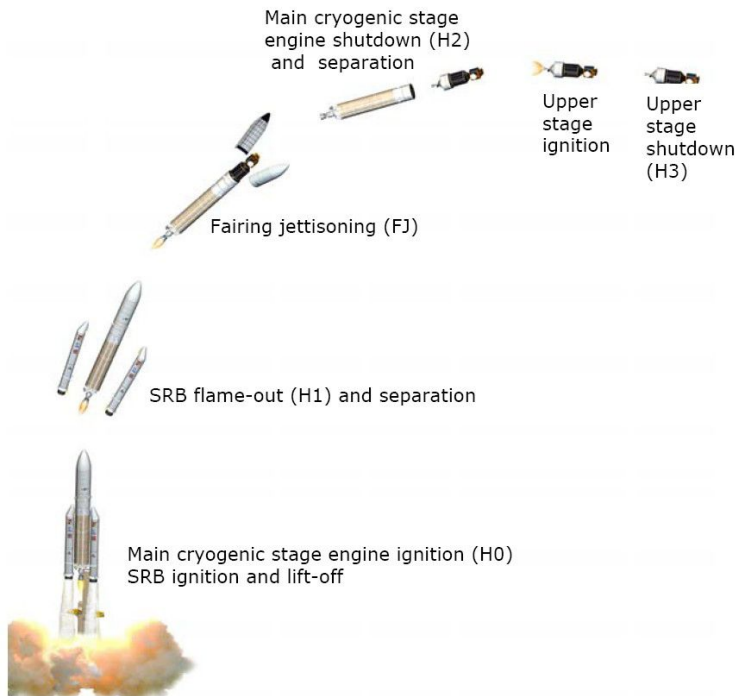


Sizing

The trend of the quasi-static acceleration from thrust of the engines for a multistage launcher, in the different flight phases, satisfies the generic relation:

$$a = \frac{T}{m} = \frac{T}{(W / g)}$$

where T is the thrust of the operational stage and W the weight of the launcher in the different phases.



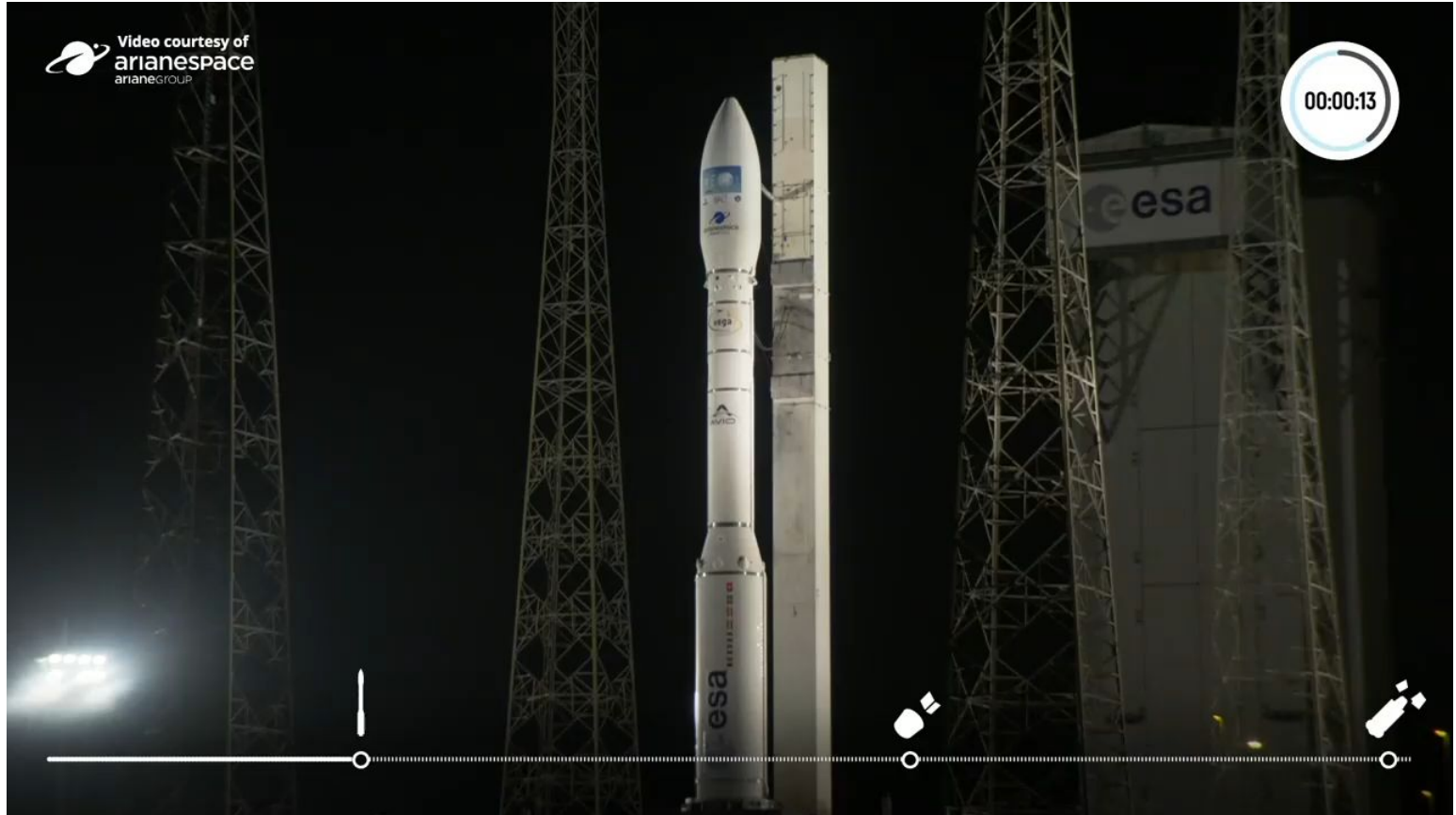
LIFT OFF!



SPACE X LAUNCH

CREDIT: SPACEX

LIFT OFF!



VEGA

CREDIT: ARIANESPACE

Quasi-static loads

Example: Quasi-Static Loads QSL specified for VEGA launcher passengers

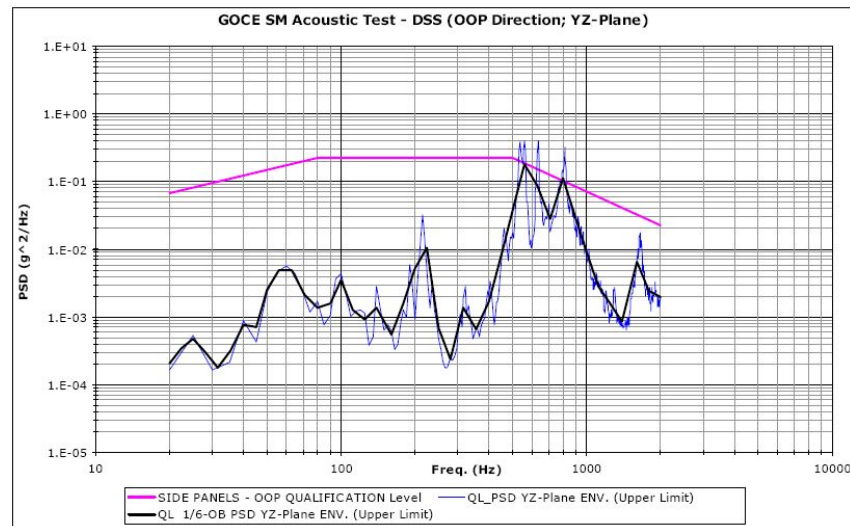
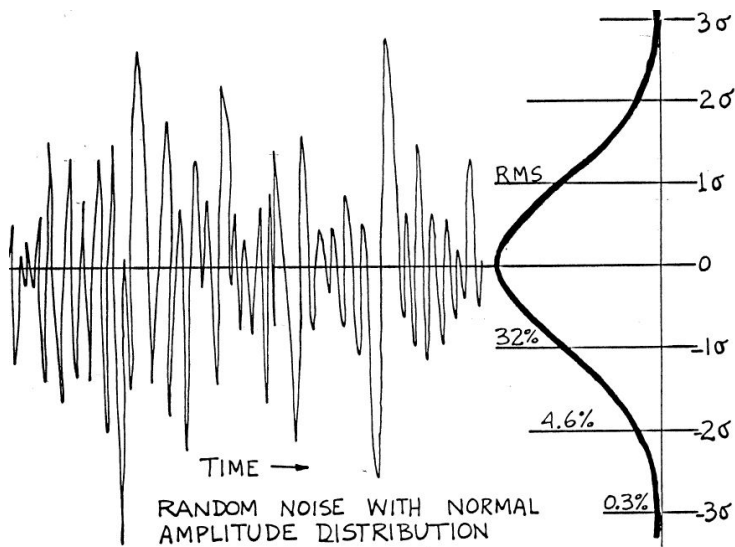
Load Event		QSL (g) (+ = tension; - = compression)					
		Longitudinal			Lateral		
		Static	Dynamic	Total	Static	Dynamic	Total
1	Lift-off phase	- 1.5	± 2.0	min - 3.5 max + 0.5	-	-	± 0.9
2	Flight with maximum dynamic pressure (Qmax)	- 2.5	± 0.5	min - 3.0 max - 2.0	-	-	± 0.9
3	First-stage flight with maximal acceleration	- 4.5	± 0.5	min - 5.0 max - 4.0	-	-	± 0.5
4	Third stage maximal acceleration	- 4.5 See note 1	± 0.2	min - 4.7 max - 4.3	-	-	± 0.2
6	Stages ignition	-	- 5.0 + 3.0	min - 5.0 max + 3.0	-	-	± 0.2

KSEC

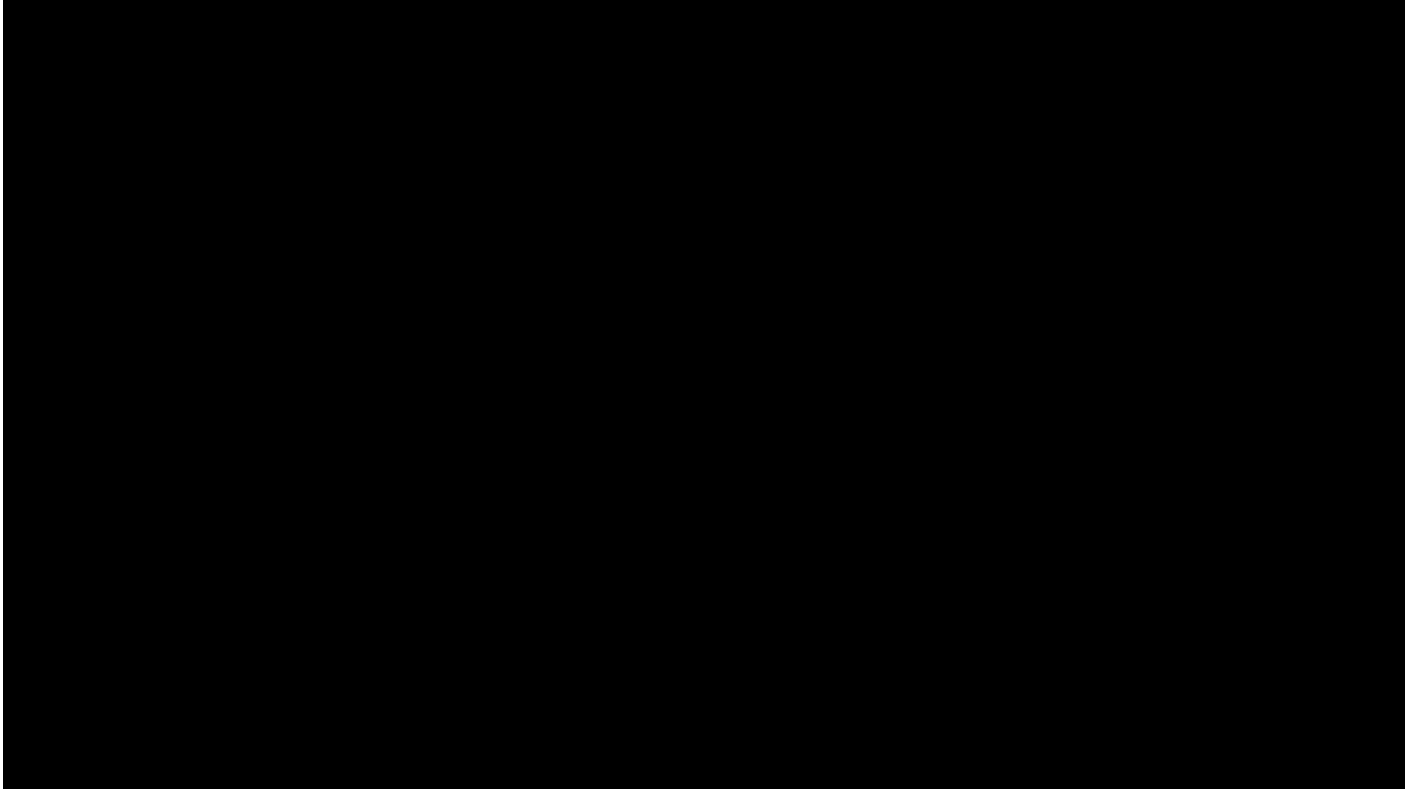


Random loads

Random loads are usually nonperiodic and their instantaneous amplitude cannot be estimated deterministically but only statistically, by means of a probability distribution (Power Spectral Density PSD). They can be caused by acoustic excitations.



Shock



Credits: [RocketLab](#)

Shock



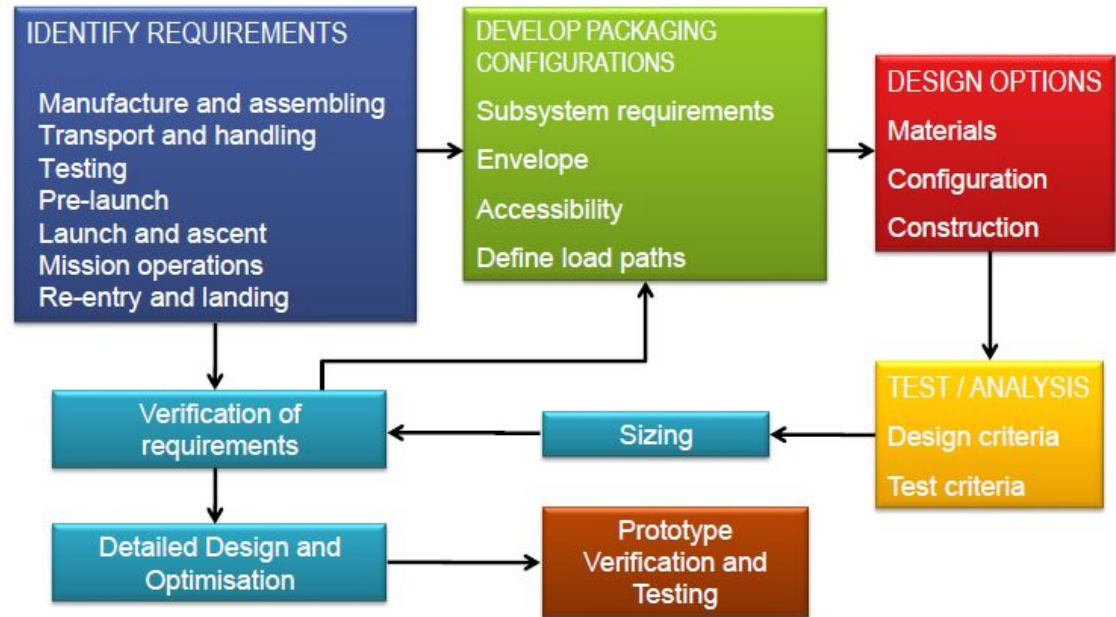
CREDIT: SPACEX

Sizing

Structural sizing process for the satellite

DESIGN: known input and desired output, determine the system

ANALYSIS: known input and the system, determine the output



Sizing: general criteria

- Structural design shall aim at **simple load paths**, maximize the use of conventional materials, simplify interfaces and **easy integration**.
- All structural assemblies and components shall be designed to **withstand applied loads due to the natural and induced environments** to which they are exposed during the service-life and shall be able, in operation, to fulfill the mission objectives for the specified duration.
- The best structure is stressed to its max allowable stress everywhere, i.e. **the best effective use of the material is obtained**. To pursue this effort and reduce the structural mass, the geometry of the structure has to be conceived in such a way to reduce the load paths and make the loads equilibrate each other in every possible location.

Sizing

Structural design (iterative process):

At the initial stage of the project

- Identification of a structural architecture
- Definition of **preliminary load factors** (at satellite level and then at lower levels)
- Preliminary structure design
- Creation of the **simplified FEM** (Finite Element Model)

Starting from phase B and in phase C/D

- Detailed definition of the structure
- FEM update and correlation with test results
- Updated environmental levels specified for lower levels
- Frequency Response Analysis (low frequencies, «sine»)
- Random vibration analysis (high frequencies)
- Shock analysis (where possible)
- Verification of satisfaction of structural requirements

Sizing

After selecting geometries, materials, constraining conditions, and carried out a preliminary sizing, we proceed with a series of checks, first analytical/numerical, then experimental.

Margin of Safety (MoS), at each point of the component, is adequately positive, for the different load conditions (yield, breakage, buckling):

$$MoS = \frac{\sigma_{allowable}}{\sigma_{applied} \cdot FoS} - 1 \geq 0$$

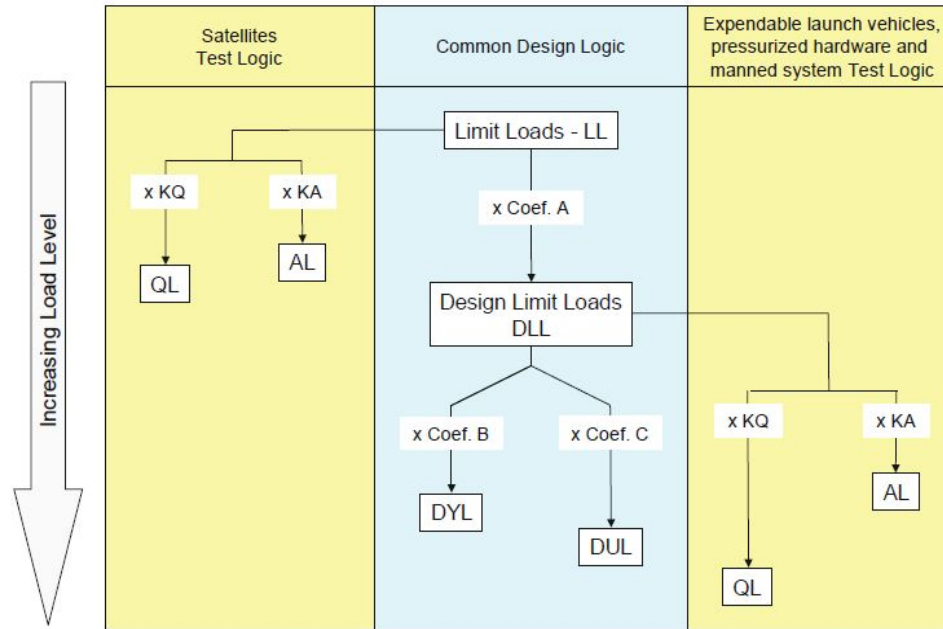
$\sigma_{allowable}$: maximum stress sustainable by the material/component without breakage, excessive deformation or damage in different load conditions (yielding - σ_Y , breaking - σ_R , eulerian buckling σ_{CR})

$\sigma_{applied}$: maximum stress actually present in the component in different loading conditions (Design Load)

(for complex stress states the reference is to an equivalent uniaxial tensile stress with a failure criterion, e.g. Von Mises).

FoS: Factor of Safety, allows uncertainties in the statistical distribution of loads, manufacturing processes, material properties and failure criteria to be taken into account.

Sizing



PROTOFLIGHT

COEFFICIENT	SATELLITE
Coeff. A or Design Factor	$KM \times KP \times KQ \times KSEC$
Coeff. B	FOSY
Coeff. C	FOSU

Typically:
 Project Factor $KP=1.1$
 Model Factor $KM=1.2$
 Qualification Factor $KQ=1.25$
 $FOSY=1.1$
 $FOSU=1.25$

Sizing

QSL from Launcher user Manual

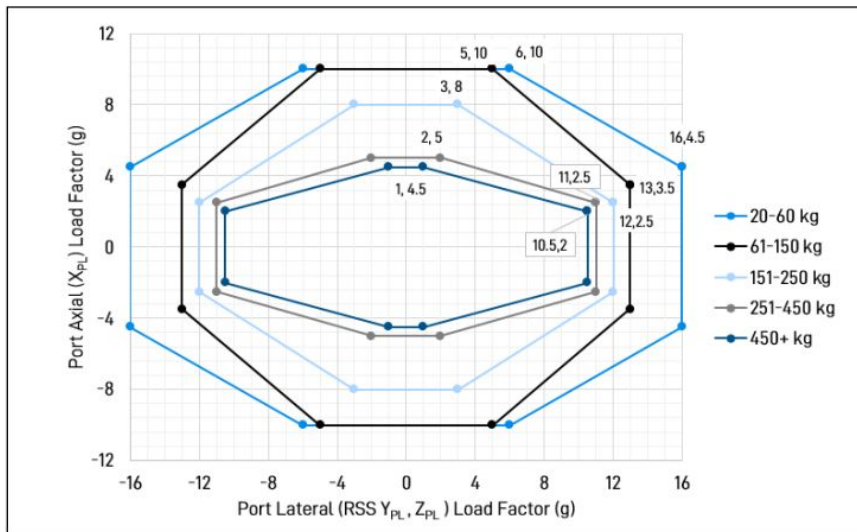


Figure 3-1: Rideshare Plate Payload Design Load Factors

Rideshare Payload Users Guide

SINE envelope

Table 3-3: Maximum Predicted Sinusoidal Vibration Environment

Frequency (Hz)	Rideshare Plate Sinusoidal Vibration MPE (g)		Starlink Adapter Sinusoidal Vibration MPE (g)	
	Axial X_{PL}	Lateral Y_{PL} , Z_{PL}	Axial X_{PL}	Lateral Y_{PL} , Z_{PL}
5	1.4	1.5	3.0	1.5
45	1.4	1.5	3.0	1.5
50	1.4	1.5	3.0	2.0
100	1.4	1.5	3.0	2.0

Sizing

Random loads from Launcher user Manual

Table 3-7: Random Vibration MPE

Frequency (Hz)	Random Vibration MPE (P95/50), All Axes
20	0.01
50	0.015
700	0.015
800	0.03
925	0.03
2000	0.00644
GRMS	5.57

Rideshare Payload Users Guide

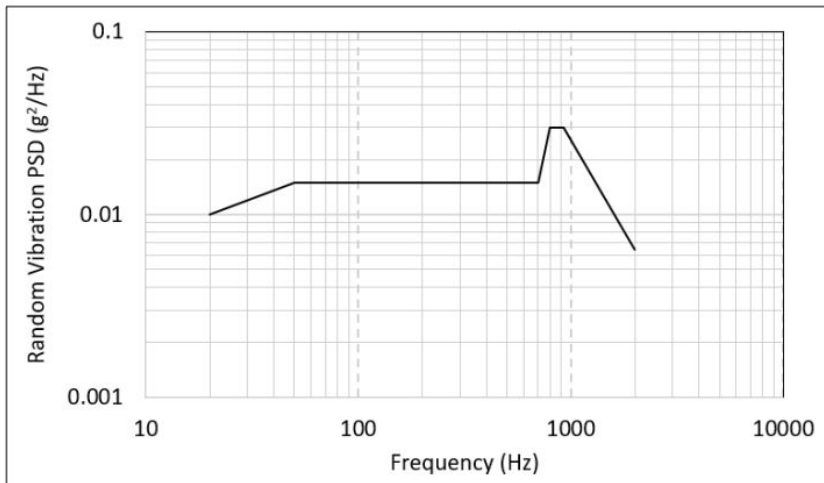


Figure 3-5: Random Vibration MPE

Sizing

- The QSLs are significant as design loads and can be assumed as per the launcher's User Manual if the launcher and satellite are decoupled from the point of view of dynamic response (excessive vibrations due to resonance conditions are not induced). This occurs if you design and constrain the satellite so that its first eigenfrequencies are above the respective launcher frequencies.
- This is defined as “sizing for stiffness”, complementary to “sizing for strength” (verification of the MoS of the structure).
- In particular, it must be verified, first analytically and then experimentally, that the frequency of the first longitudinal and lateral modes of the satellite are greater than the limits specified in the launcher's User Manual ($f_{SC} > 1,4 f_{LV}$).
- For components, it must be verified that the first natural frequency of the component > 2 first natural frequency of the support

Example: frequency requirements for Space X Rideshare mission

FIRST NATURAL FREQUENCIES	LAUNCH VEHICLE SPECIFICATION
LONGITUDINAL AXIS	$f \geq 40 \text{ Hz}$
LATERAL AXIS	$f \geq 40 \text{ Hz}$

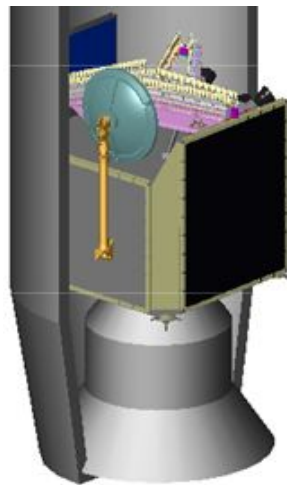
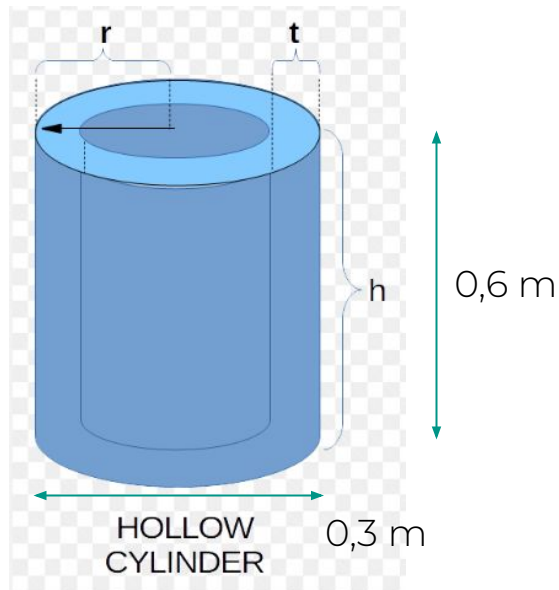
Typical requirement for CubeSats: $> 90 \text{ Hz}$

Structure sizing

GOAL: size the structure of a microsatellite structure

Assumption: we approximate the structure of the satellite as a hollow cylinder (monocoque configuration: easy approach for preliminary analysis)

The process is iterative!



- Mass: 50 kg
- Material: Al7075
- $\rho=2800 \text{ kg/m}^3$
- $\sigma_y=350 \text{ MPa}$
- $\sigma_u=500 \text{ MPa}$
- $E = 70 \text{ GPa}$

STATIC REQUIREMENTS

Axial Acc.= 15 g

Lateral Acc.= 15 g

DYNAMIC REQUIREMENTS

$F_{nat_Ax} = 100 \text{ Hz}$

$F_{lat_Ax} = 100 \text{ Hz}$

Structure sizing

1) DYNAMIC SIZING



$$a) \quad f_{ax} = 0,25 \sqrt{\frac{AE}{mL}} \quad \square \quad 100 = 0,25 \sqrt{\frac{A \cdot 70 \cdot 10^9}{50 \cdot 0,6}} \quad \square \quad \text{Solve for A}$$

$$\square t = 0,073 \text{ mm}$$

We use this first value of thickness to calculate f_{lat}

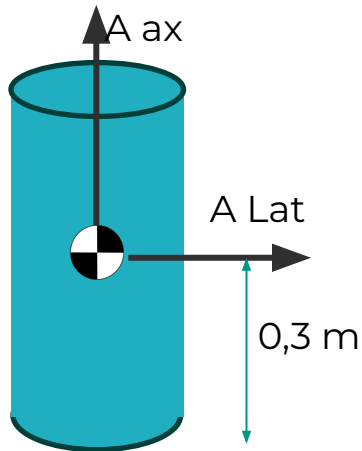
$$b) \quad f_{lat} = 0,56 \sqrt{\frac{EI}{mL^3}} \quad \square \quad \text{Use preliminary thickness to calculate I}$$

$$\square I = \pi(D^4 - d^4)/64 \quad \square f_{lat} = 39 \text{ Hz} \quad \text{Too Low!}$$

In order to meet the requirement the thickness shall be 0,48 mm

Structure sizing

2) STATIC SIZING



$$F_{Ax} = mA_{ax} \quad \square \quad F_{Ax} = 7357,5 \text{ N}$$

$$F_{Lat} = mA_{Lat} \quad \square \quad F_{Lat} = 7357,5 \text{ N}$$

To calculate the bending moment we multiply F_{Lat} by the distance between the CoG and the boundary

$$\square \quad M = F_{Lat} h/2 \quad \square \quad M = 2207 \text{ Nm}$$

Calculate σ and τ

$$\sigma = \frac{F_{ax}}{A} + M \frac{R}{I} \quad \square \quad \sigma_{vm} = \sqrt{\sigma^2 + 3\tau^2} \quad \square \quad \sigma_{vm} = 110 \text{ MPa}$$

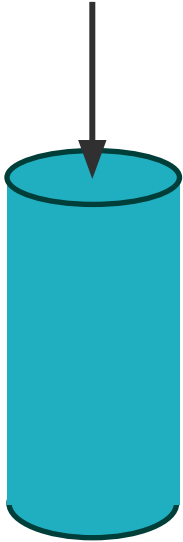
$$\tau = \frac{F_{lat}}{A}$$

Calculate MOS

$$MOS = \frac{\sigma_{amm}}{FOSy \sigma_{vm}} - 1 \quad \square \quad MOS = 1,7 \quad \text{MOS is positive !!}$$

Structure sizing

3) INSTABILITY SIZING



$$P_{cr} = A\sigma_{cr} = \frac{\pi^2 EI}{4L^2}$$

For a hollow cylinder the formula is

$$\sigma_{cr} = 0,6\gamma \frac{Et}{R}$$

Where γ is a parameter dependent on geometry

By applying the formula $\square P_{cr} = 22900 \text{ N}$

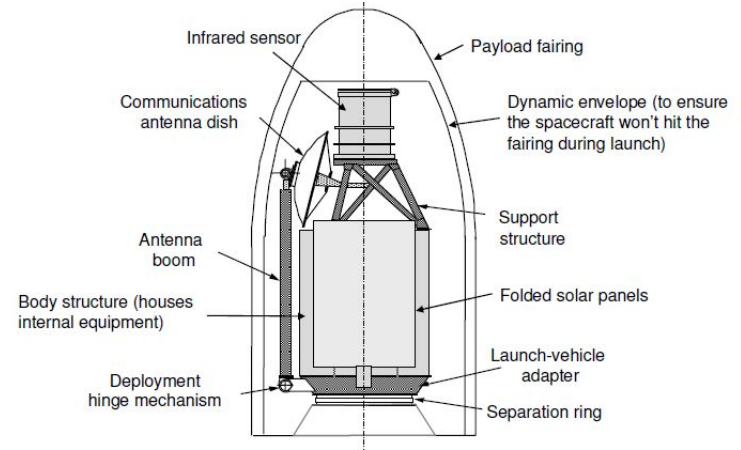
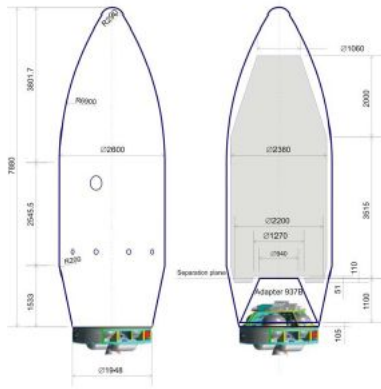
$F_{ax} < P_{cr} ! \text{ OK}$

- Actually, depending on the thickness of the structure the mass of the satellite would change and so there is an iteration.
- Structure accounts for 10 – 20% for a microsatellite

Sizing

...Other aspects to consider

Often, along with purely structural aspects, other aspects related to satellite configuration need to be addressed. Subsystems must be accommodated so that they can operate at their best (unobstructed fields of view, unobstructed surfaces, mutual compatibility).



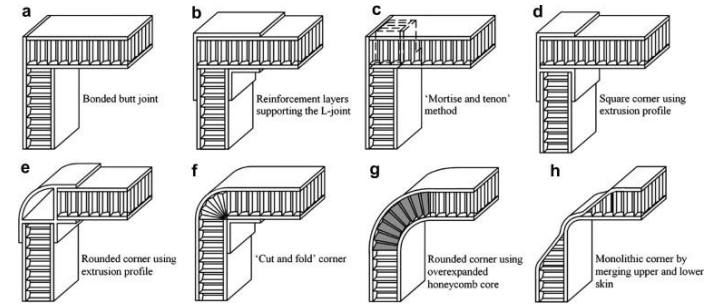
The total volume, in the "stowed" configuration must be compatible with that available in the fairing of the selected launcher.

Materials

The components of the **Structure subsystem** consist of structural elements, whose geometry and materials arise from the described (iterative) design process.

A critical step is the appropriate **choice of materials** for the specific application; the main parameters considered are:

- Strength (allowable loads)
- Stiffness (Young's modulus)
- Density,
- Resistance to corrosion
- Thermal properties (CTE),
- Ease of processing
- Cost
- Availability.



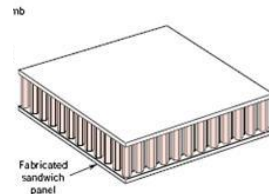
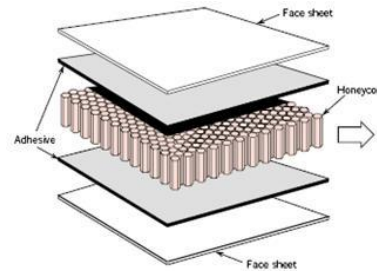
SANDWICHES CONSTRAINT METHODS

These material characteristics must be reproducible but are often known with some uncertainty.

The choice often falls between **metal alloys** (aluminium, titanium) and **composite materials** (carbon fiber laminates Carbon Fiber Reinforced Plastic CFRP, sometimes combined in sandwiches with an aluminum honeycomb core) held together by inserts.

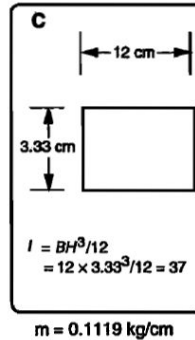
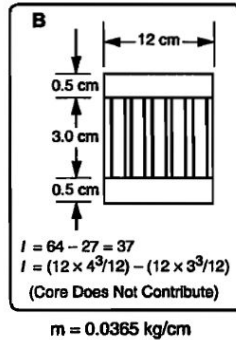
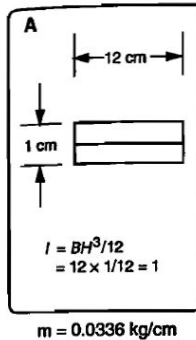
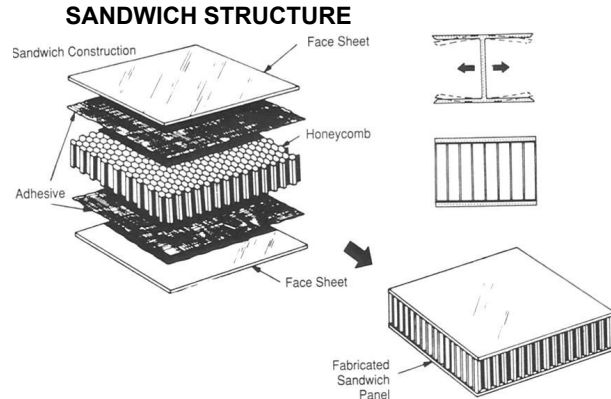
Materials

Characteristics of the most used materials for space constructions

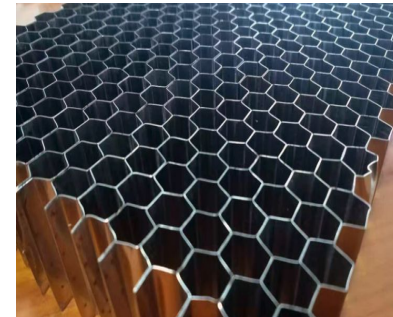
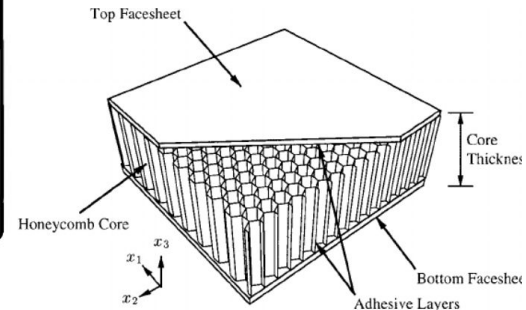
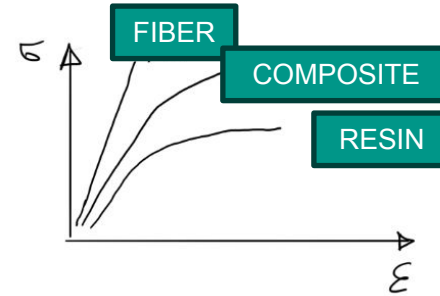


Material	Advantages	Disadvantages
<i>Aluminum</i>	• High strength vs. weight • Ductile; tolerant of concentrated stresses • Easy to machine • Low density; efficient in compression	• Relatively low strength vs. volume • Low hardness • High coefficient of thermal expansion
<i>Steel</i>	• High strength • Wide range of strength, hardness, and ductility obtained by treatment	• Not efficient for stability (high density) • Most are hard to machine • Magnetic
<i>Heat-resistant</i>	• High strength vs. volume • Strength retained at high temperatures • Ductile	• Not efficient for stability (high density) • Not as hard as some steels
<i>Magnesium</i>	• Low density—very efficient for stability	• Susceptible to corrosion • Low strength vs. volume
<i>Titanium</i>	• High strength vs. weight • Low coefficient of thermal expansion	• Hard to machine • Poor fracture toughness if solution treated and aged
<i>Beryllium</i>	• High stiffness vs. density	• Low ductility & fracture toughness • Low short transverse properties • Toxic
<i>Composite</i>	• Can be tailored for high stiffness, high strength, and extremely low coefficient of thermal expansion • Low density • Good in tension (e.g., pressurized tanks)	• Costly for low production volume; requires development program • Strength depends on workmanship; usually requires individual proof testing • Laminated composites are not as strong in compression • Brittle; can be hard to attach

Materials

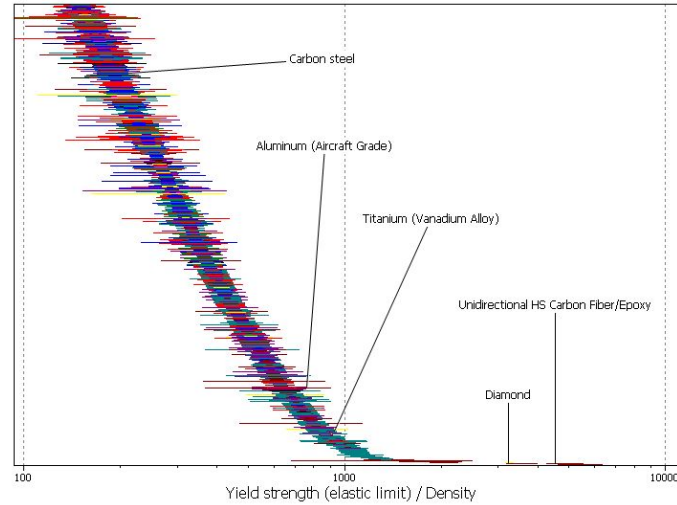
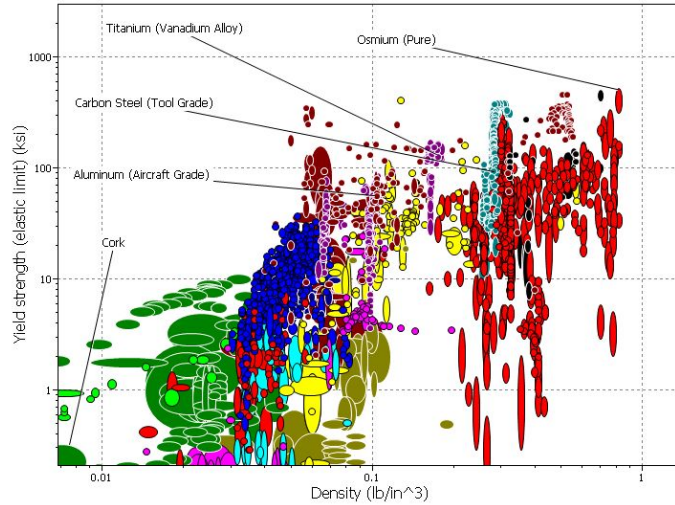


COMPOSITE CHARACTERISTIC CURVES
FIBER + RESIN



Materials

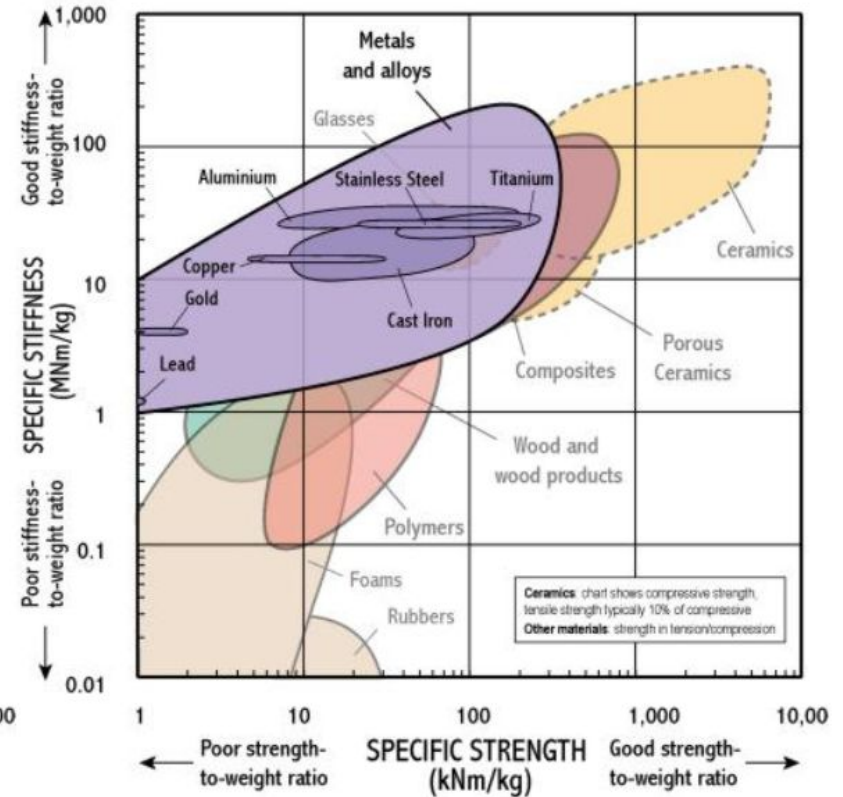
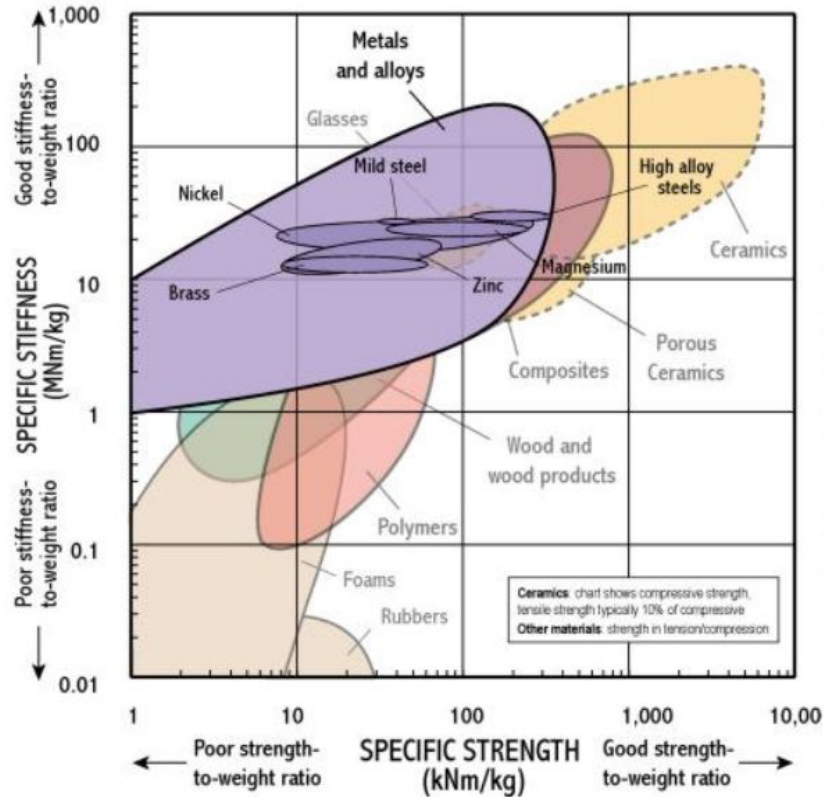
SPECIFIC RESISTANCE



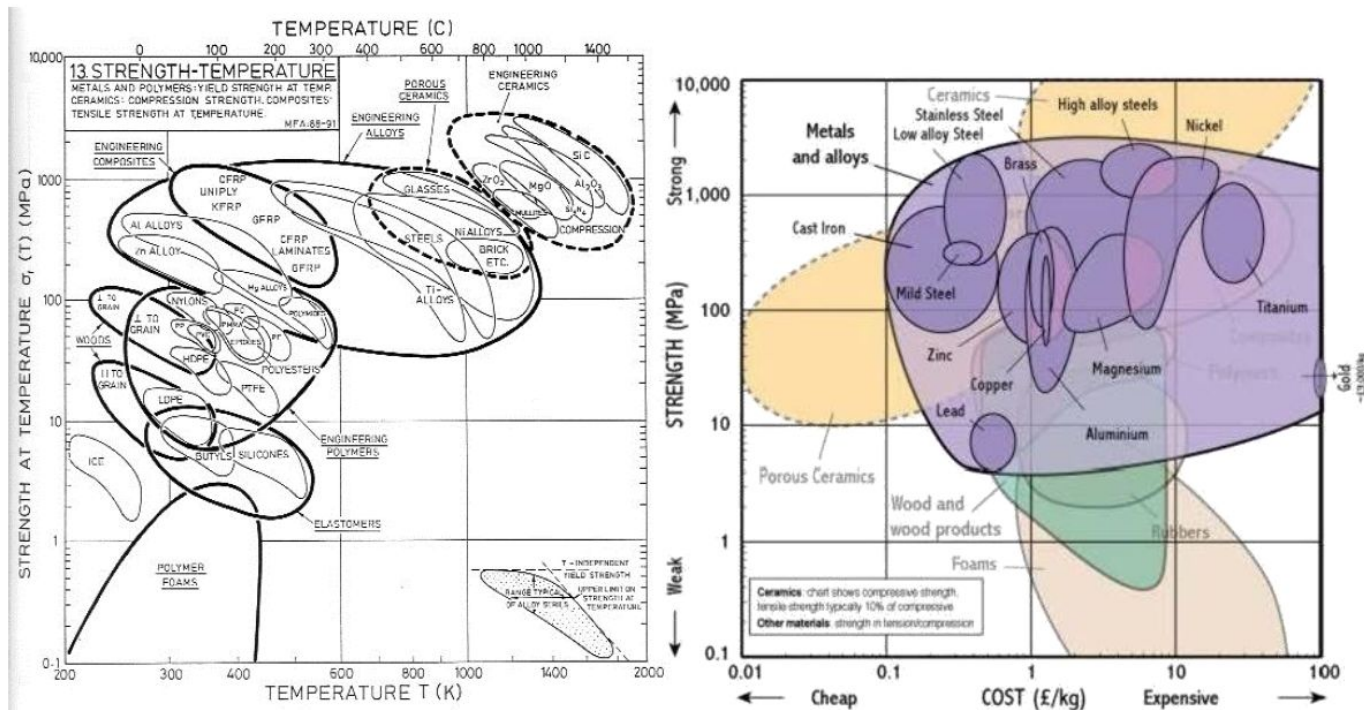
Materials

SPECIFIC STIFFNESS AND STRENGTH

Metals and alloys



TEMPERATURE RESISTANCE



Materials

Structural parameters of the most used materials for space structures

Material	Density (kg/m ³)	Y.M. (GPa)	Yield (MPa)	Thermal Expansion (10 ⁻⁶ /K)
Al	2800	68	276	23.6
Mg	1700	71	503	23.4
Ti	4400	110	825	9
Be	2000	304	345	11.5
Composite CFRP	1620	281	586	[-11.7,29.7]

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